

DA 5001 / DA 6336: Privacy in AI

Krishna Pillutla

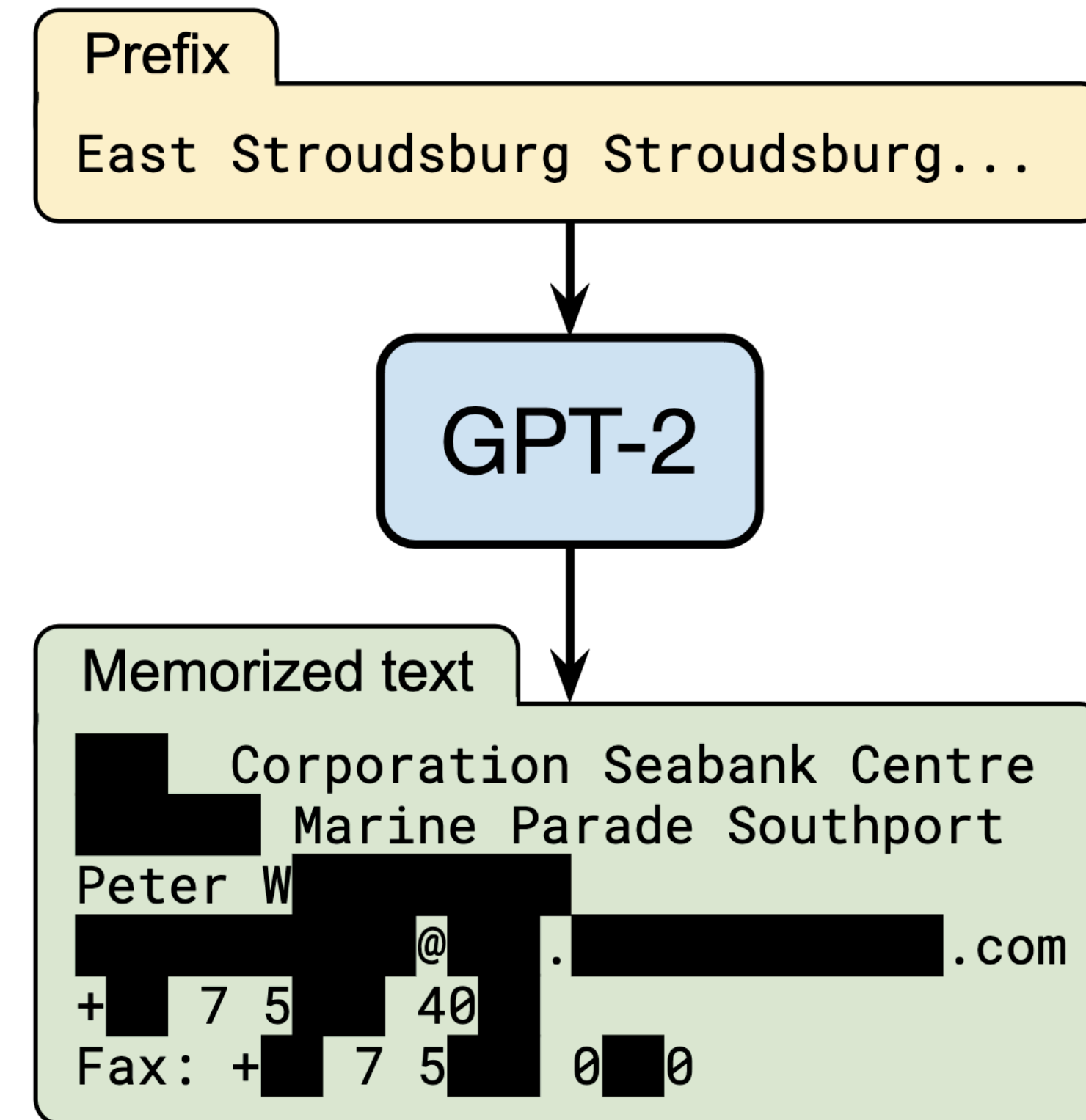
IIT Madras



Models leak information about their training data

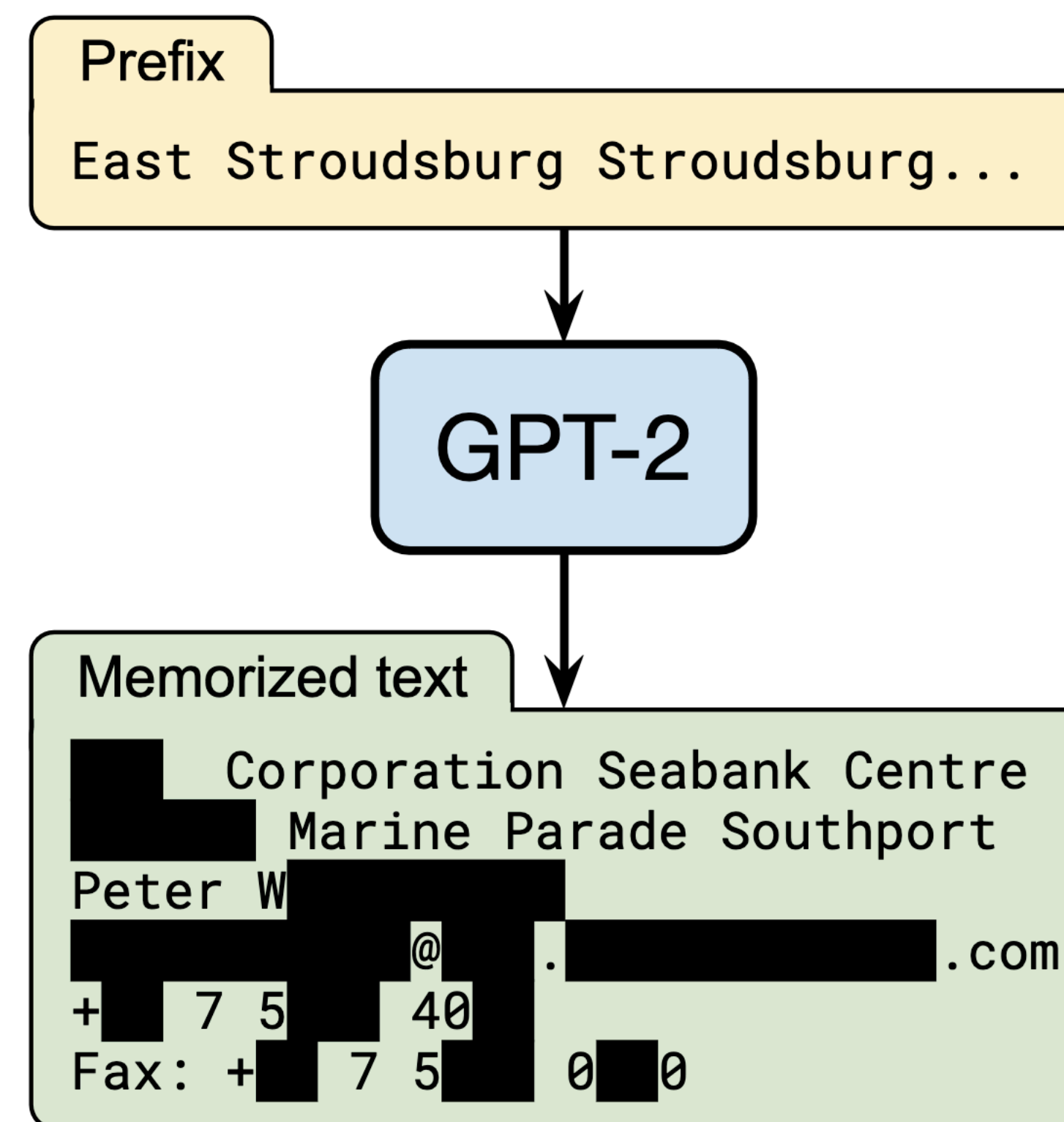
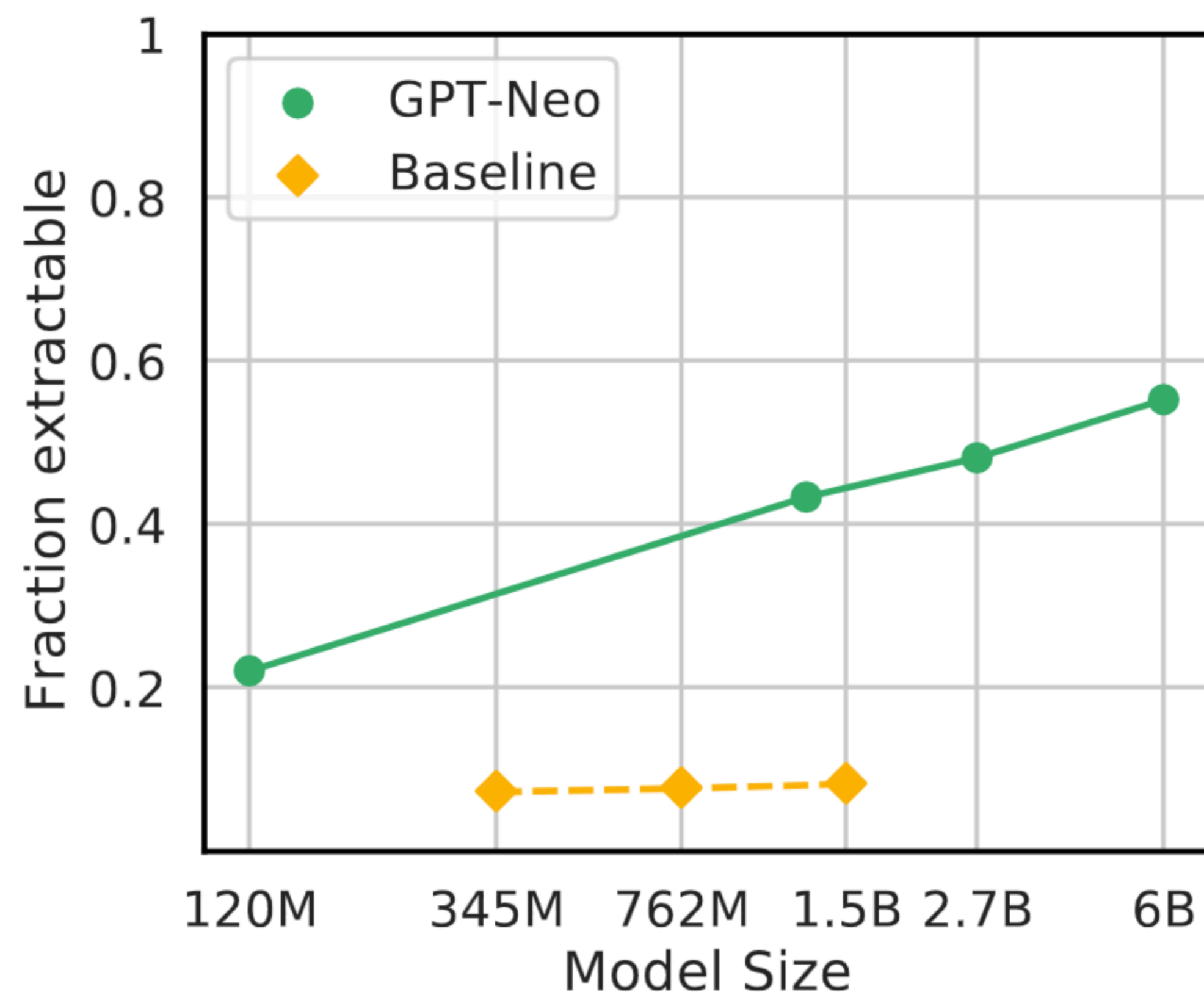


WHEN YOU TRAIN PREDICTIVE MODELS
ON INPUT FROM YOUR USERS, IT CAN
LEAK INFORMATION IN UNEXPECTED WAYS.



Carlini et al. (USENIX Security 2021)

Models leak information about their training data *reliably*



Carlini et al. (USENIX Security 2021)

Diffusion Art or Digital Forgery? Investigating Data Replication in Diffusion Models

Gowthami Somepalli 🐢, Vasu Singla 🐢, Micah Goldblum 🗽, Jonas Geiping 🐢, Tom Goldstein 🐢

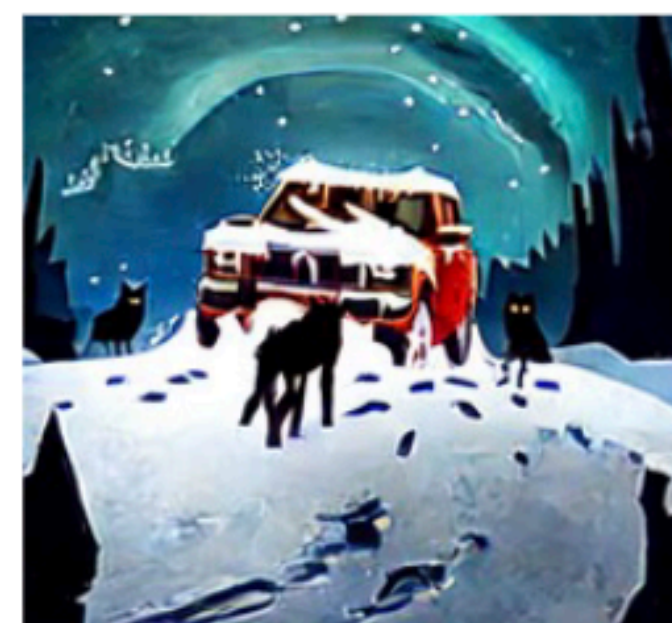
🐢 University of Maryland, College Park

{gowthami, vsingla, jgeiping, tomg}@cs.umd.edu

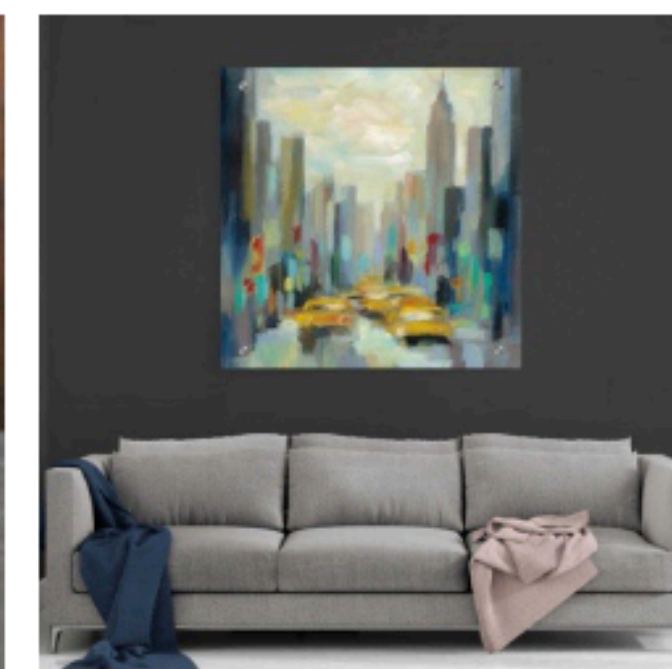
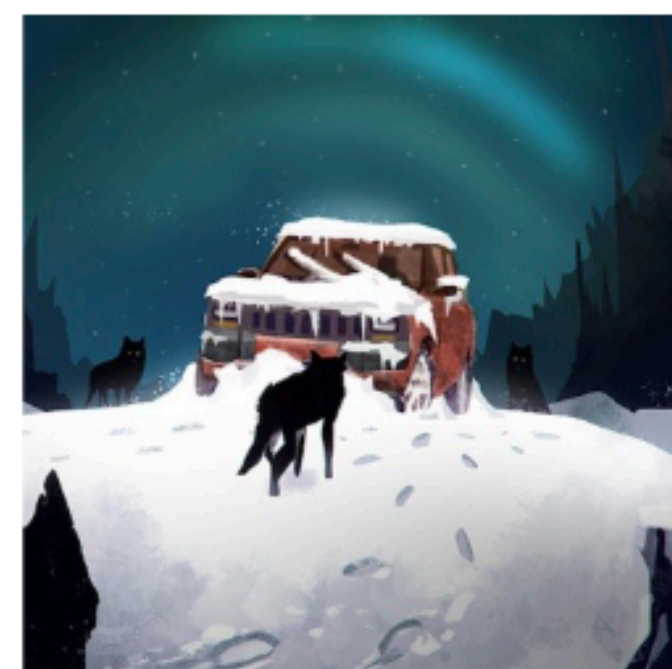
🗽 New York University

goldblum@nyu.edu

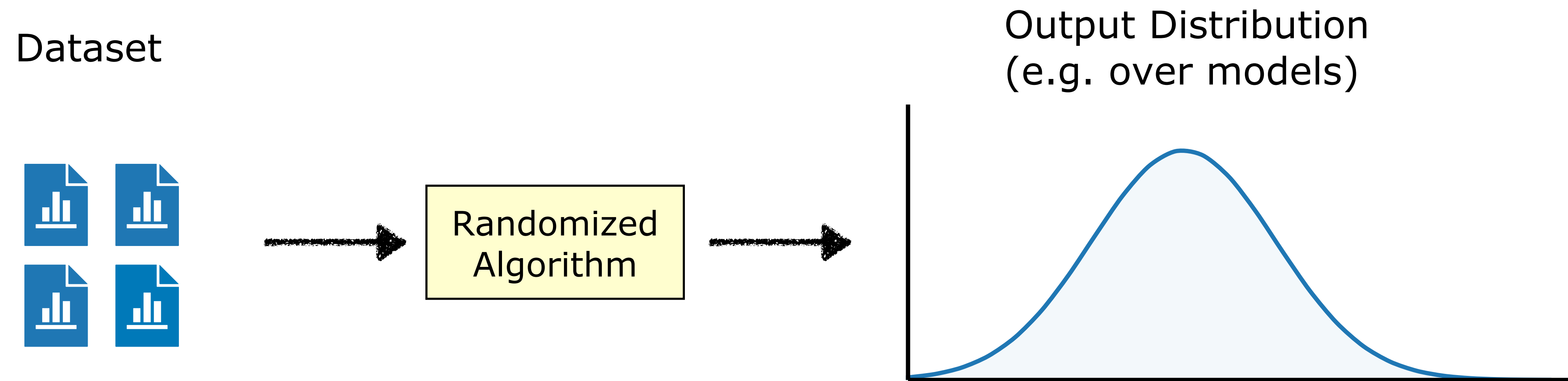
Generation



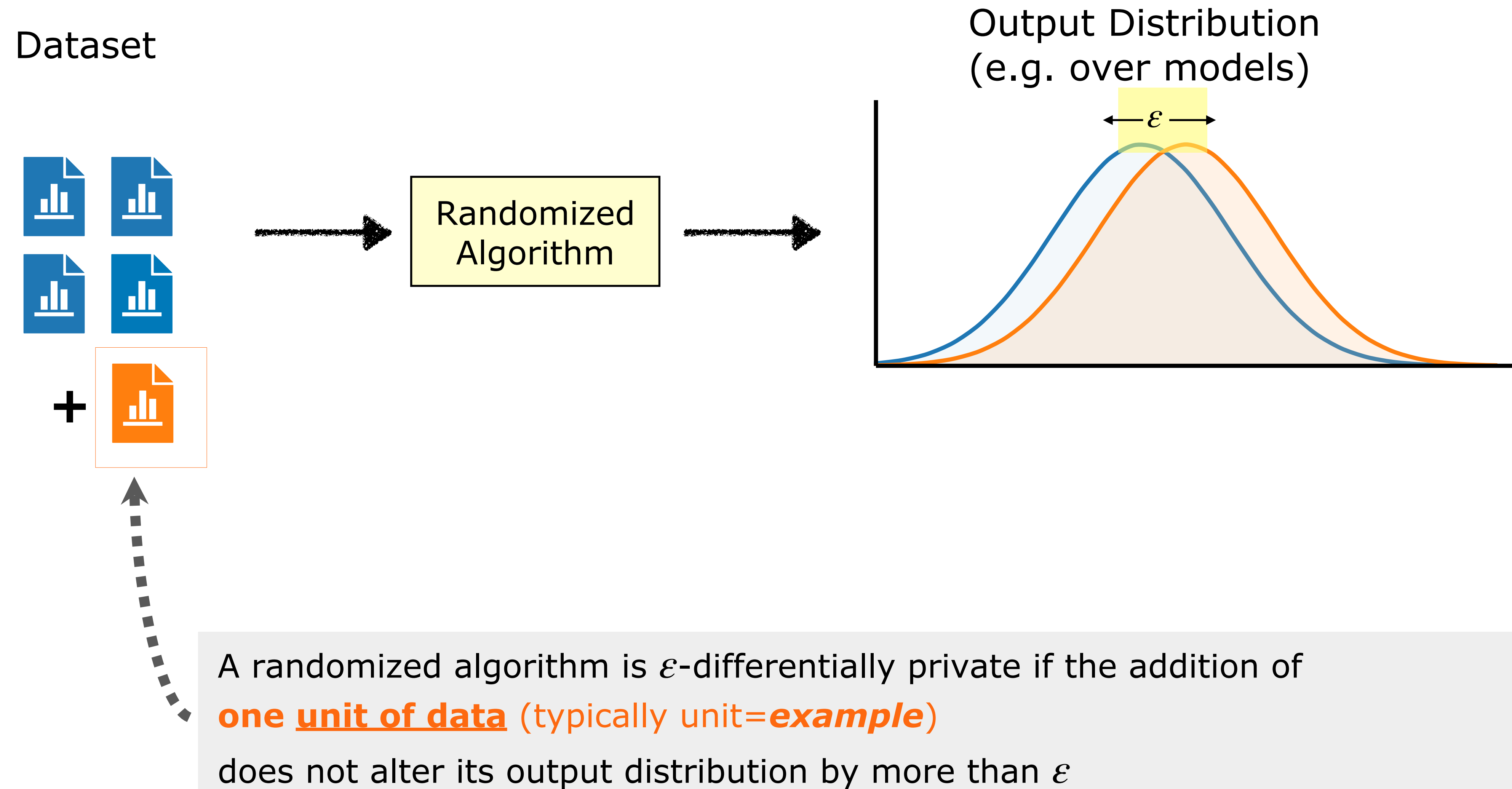
LAION-A Match



Differential privacy (DP)

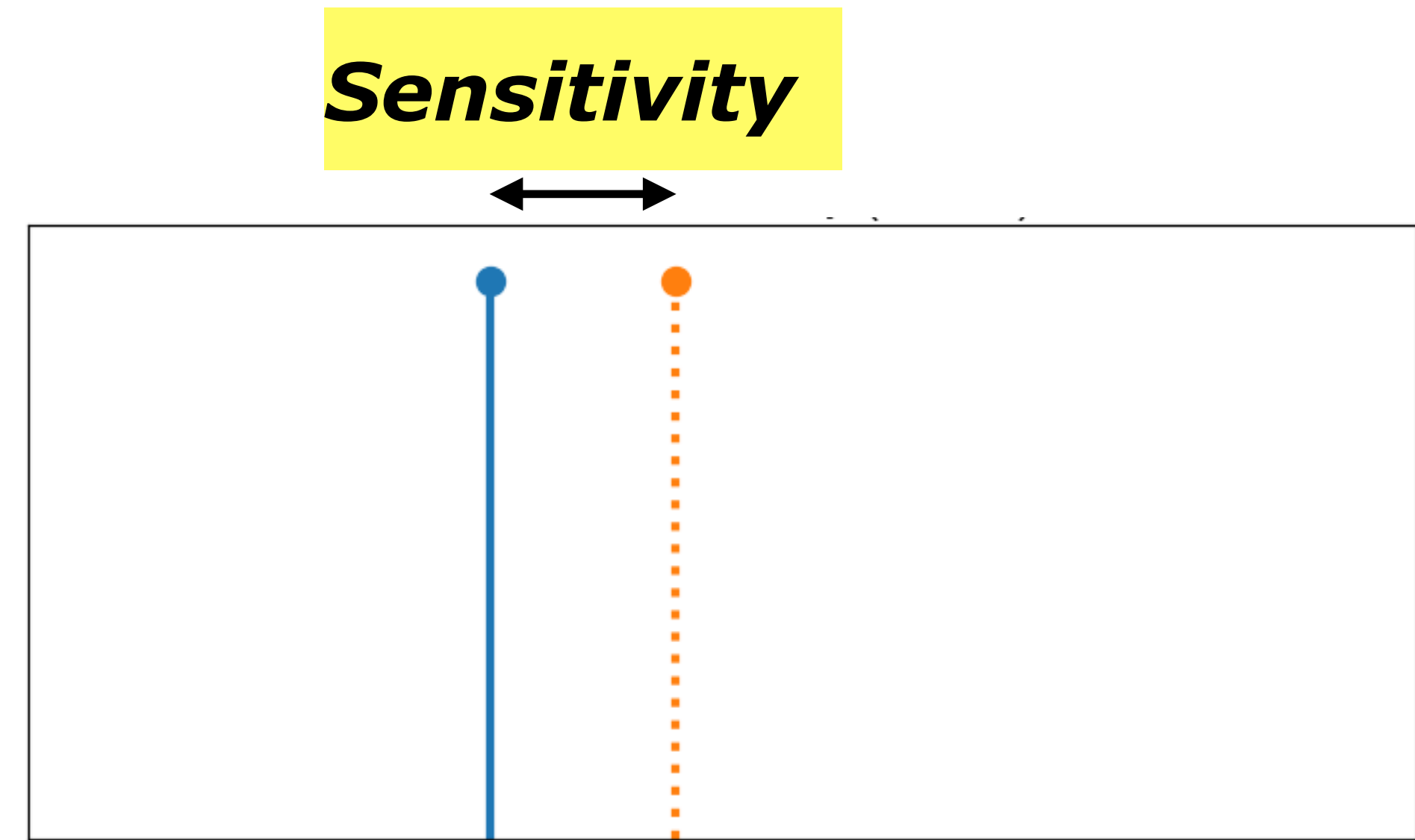
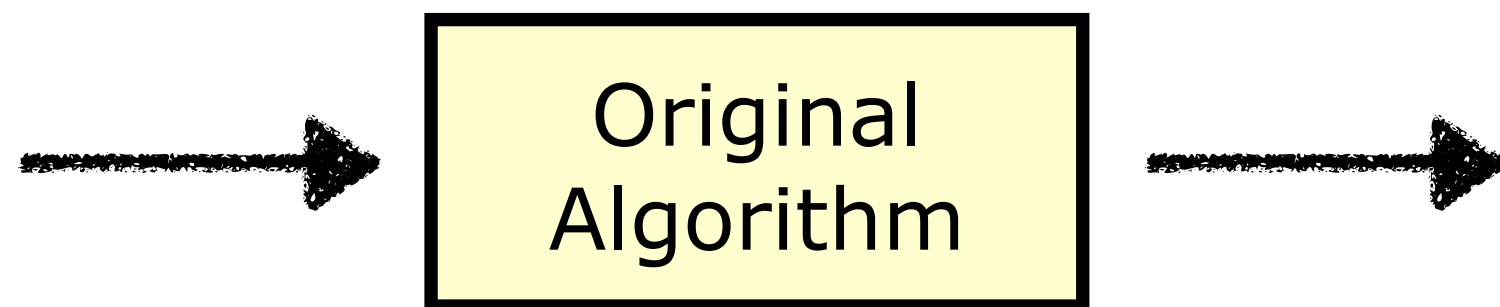


Differential privacy (DP)



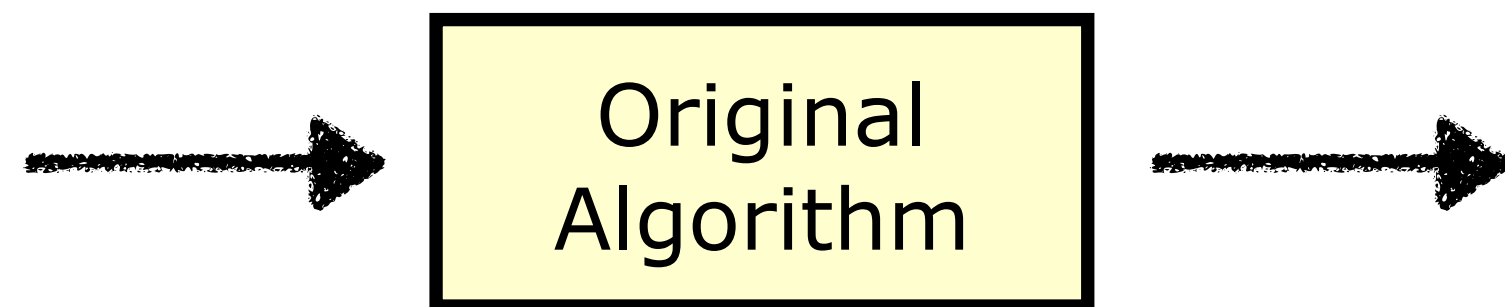
Adding noise for DP

Dataset



Adding noise for DP

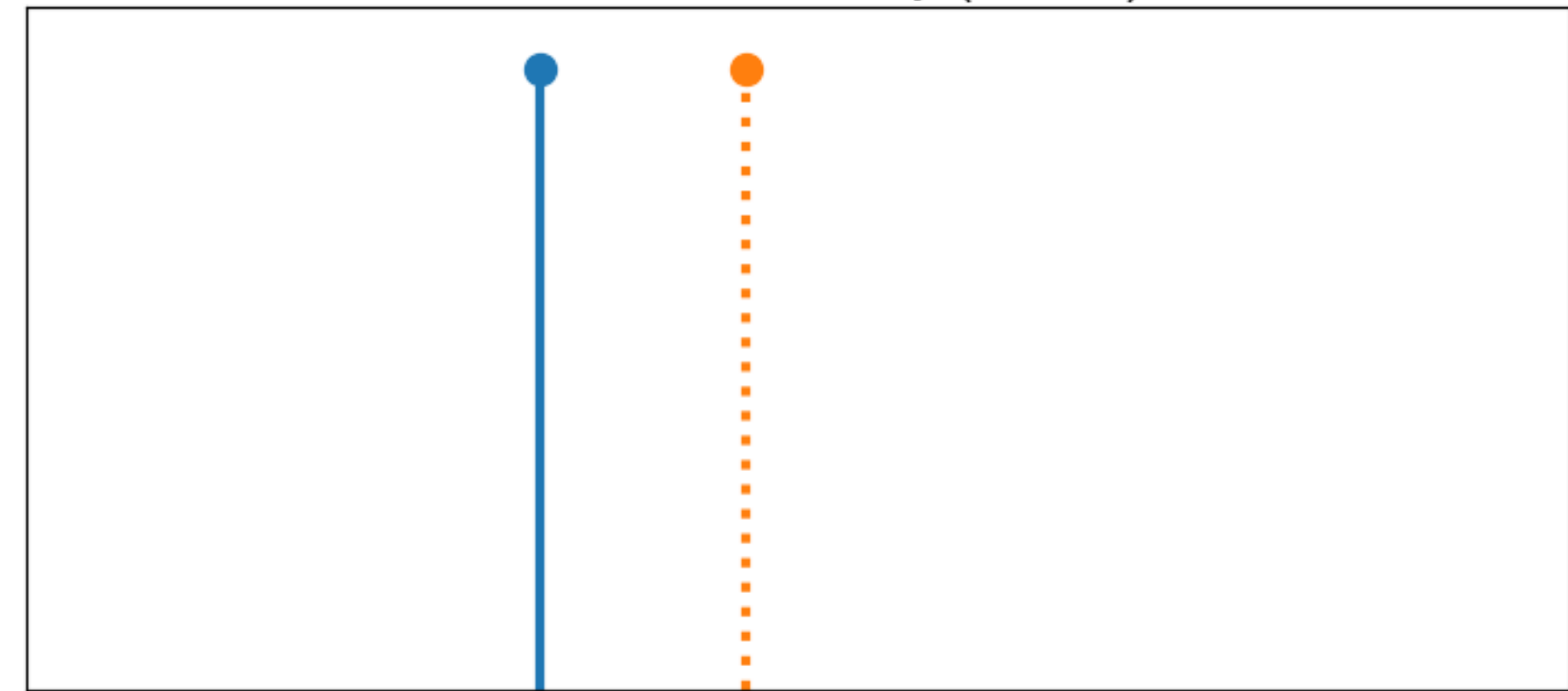
Dataset



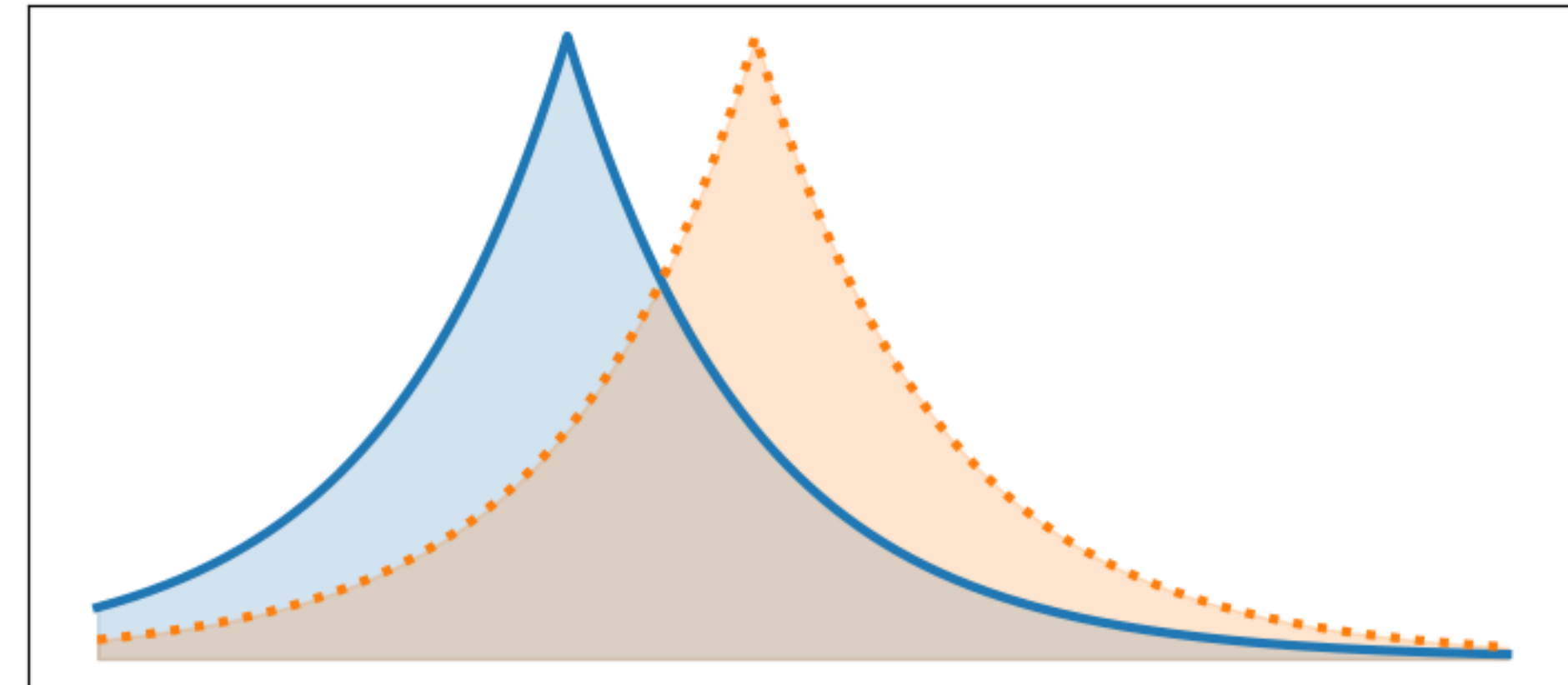
Add Noise
(e.g. Laplace/Gaussian)



Sensitivity

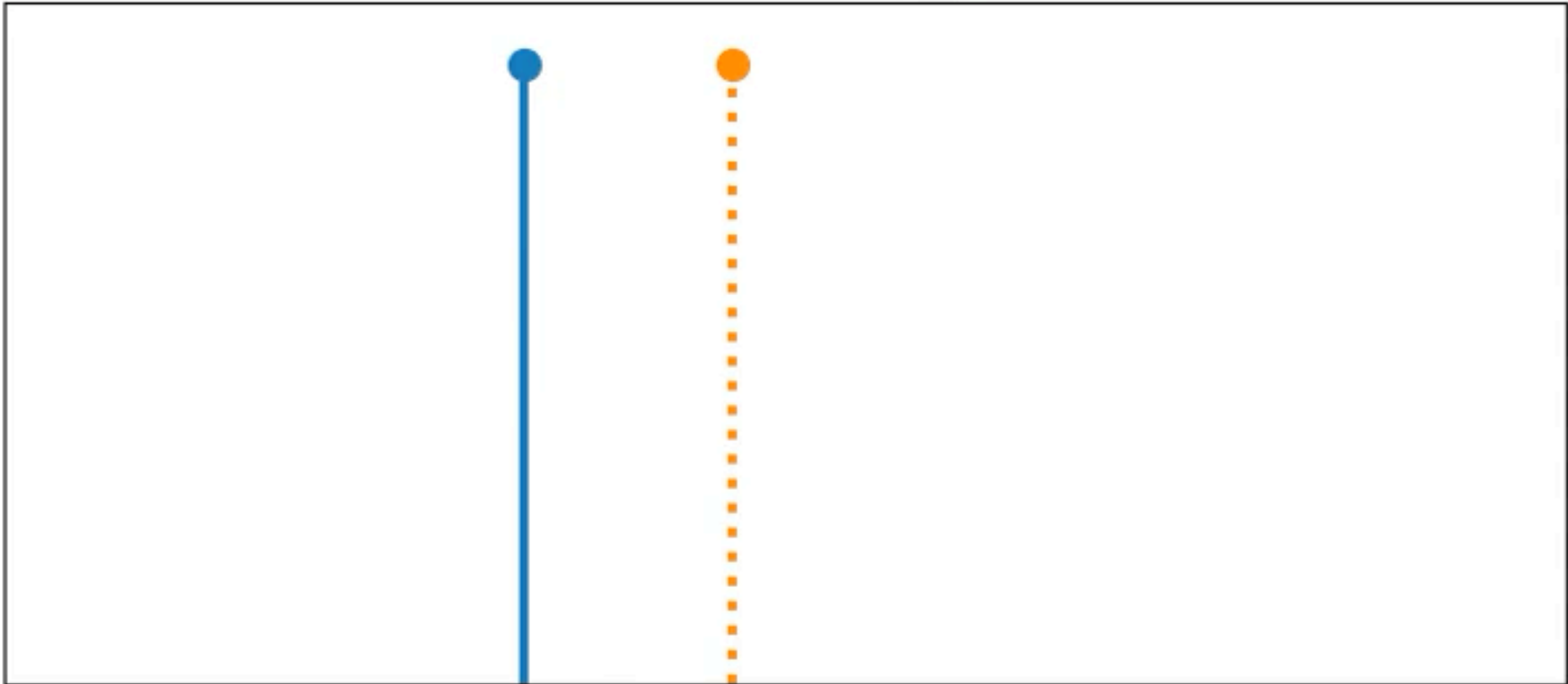


Differential Privacy with $\epsilon = 1$

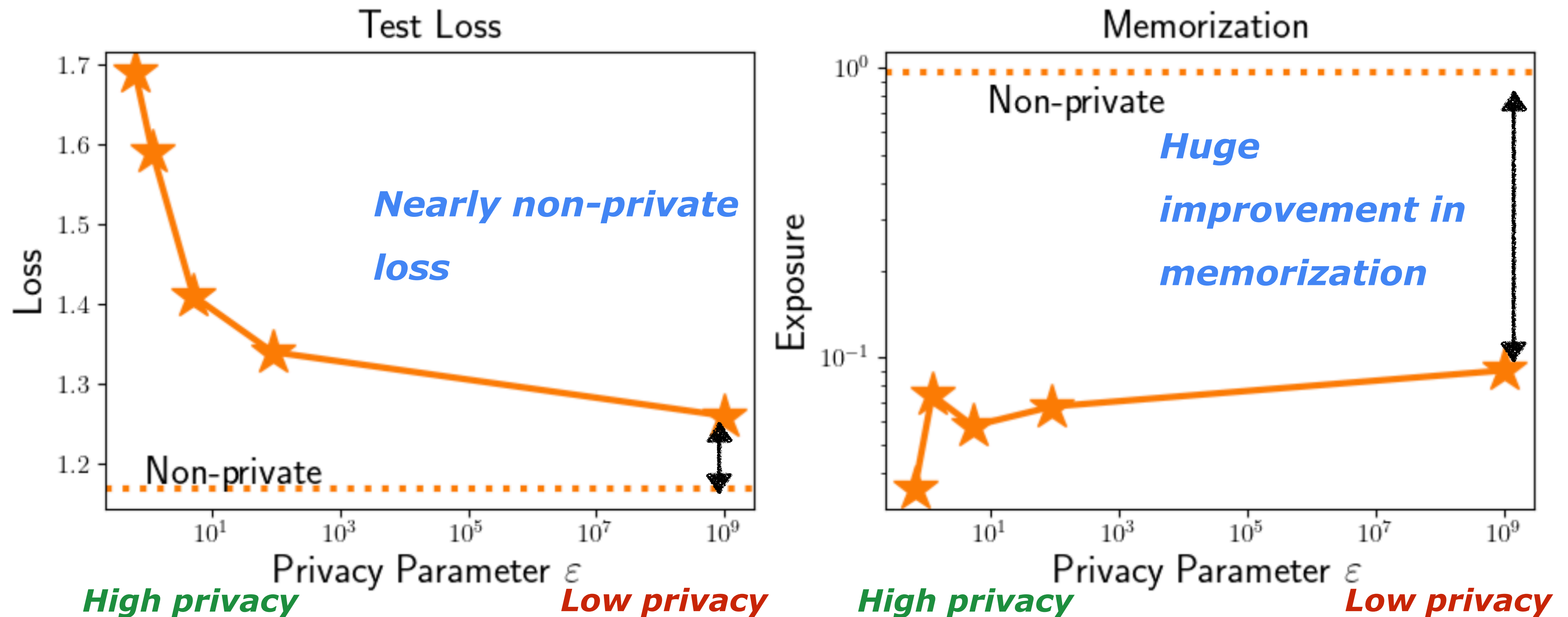


Adding noise for DP

No Differential Privacy ($\epsilon = \infty$)

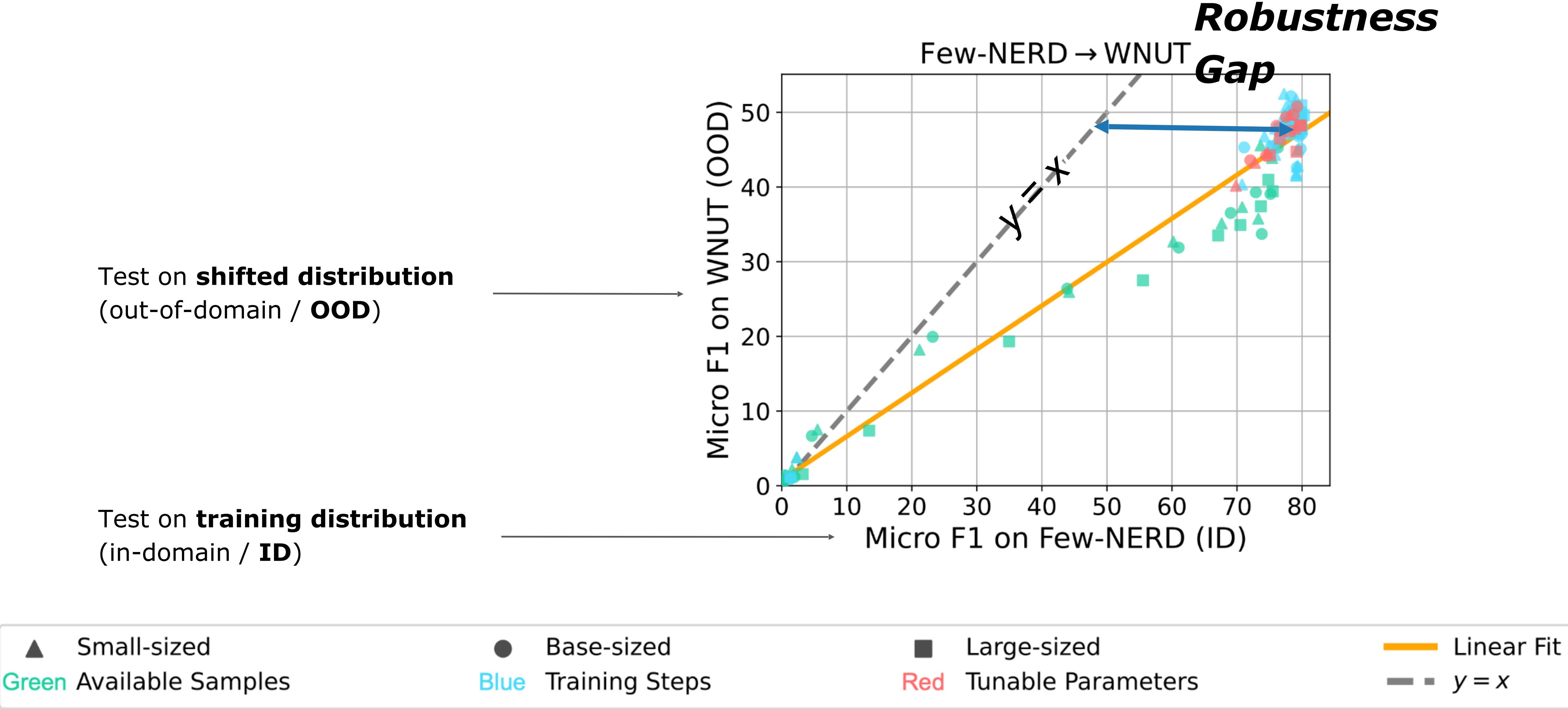


Training with differential privacy nearly eliminates memorization



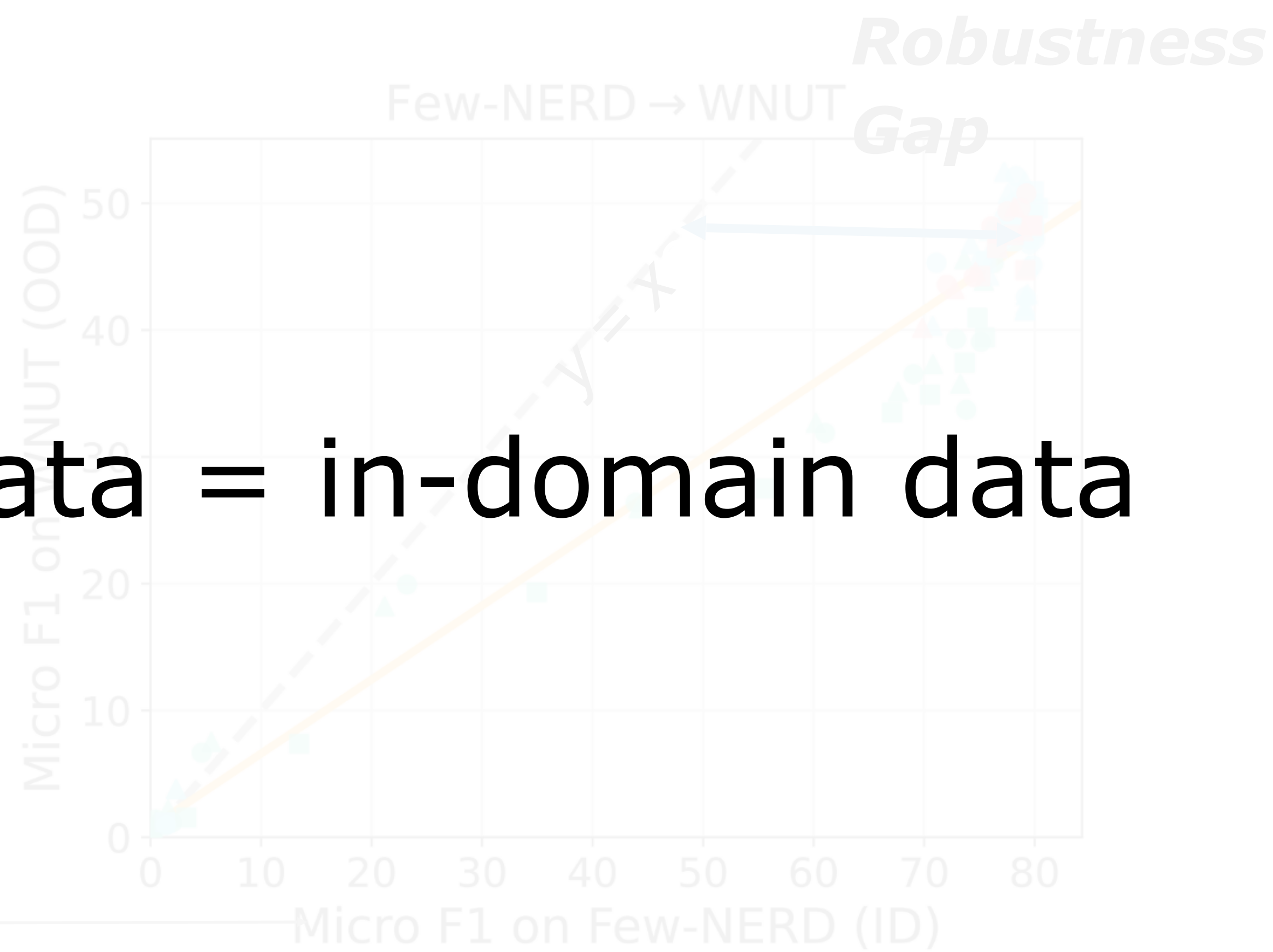
Carlini, Liu, Erlingsson, Kos, Song. **The Secret Sharer: Evaluating and Testing Unintended Memorization in Neural Networks.** USENIX Security 2019.

Why even use sensitive data? Why not off-the-shelf LLMs?



Which data do we use to train/finetune/align these models?

Best training data = in-domain data



▲ Small-sized

● Base-sized

■ Large-sized

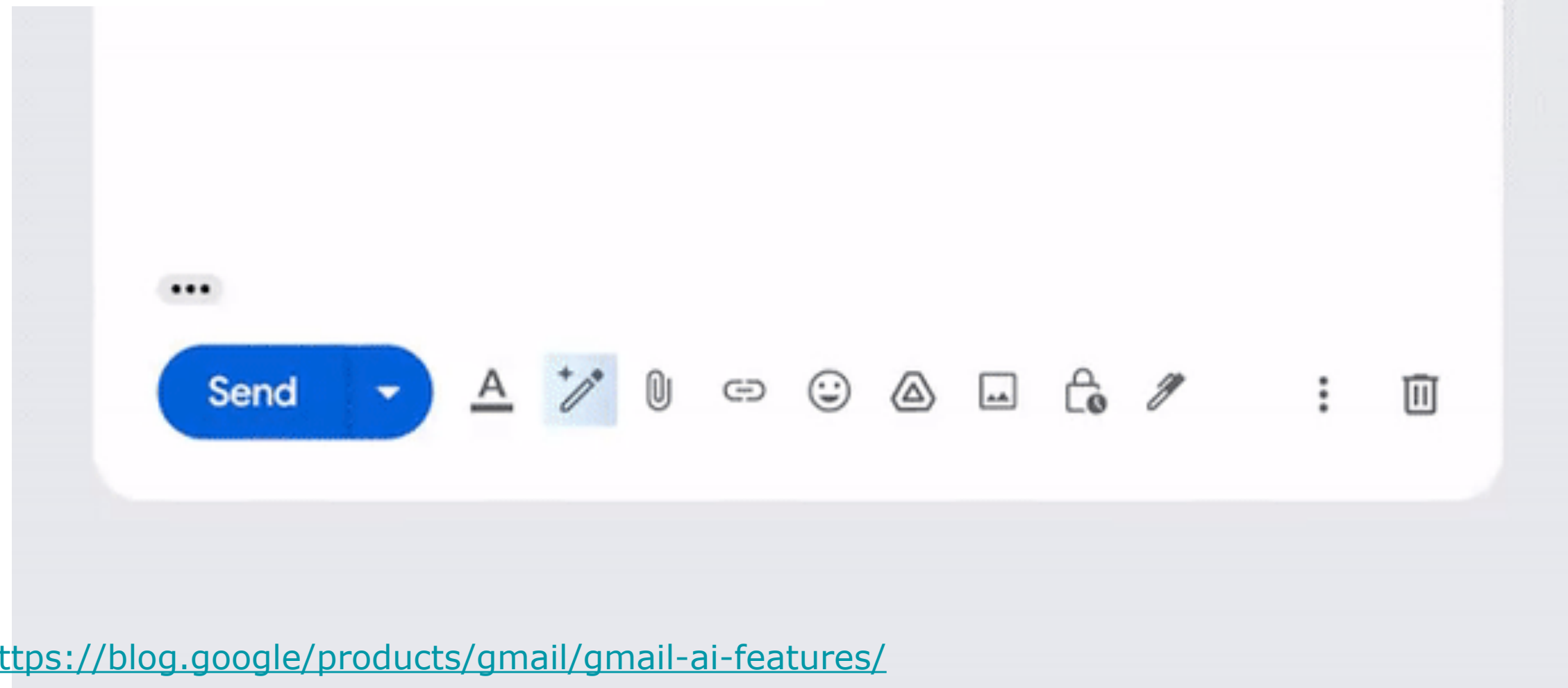
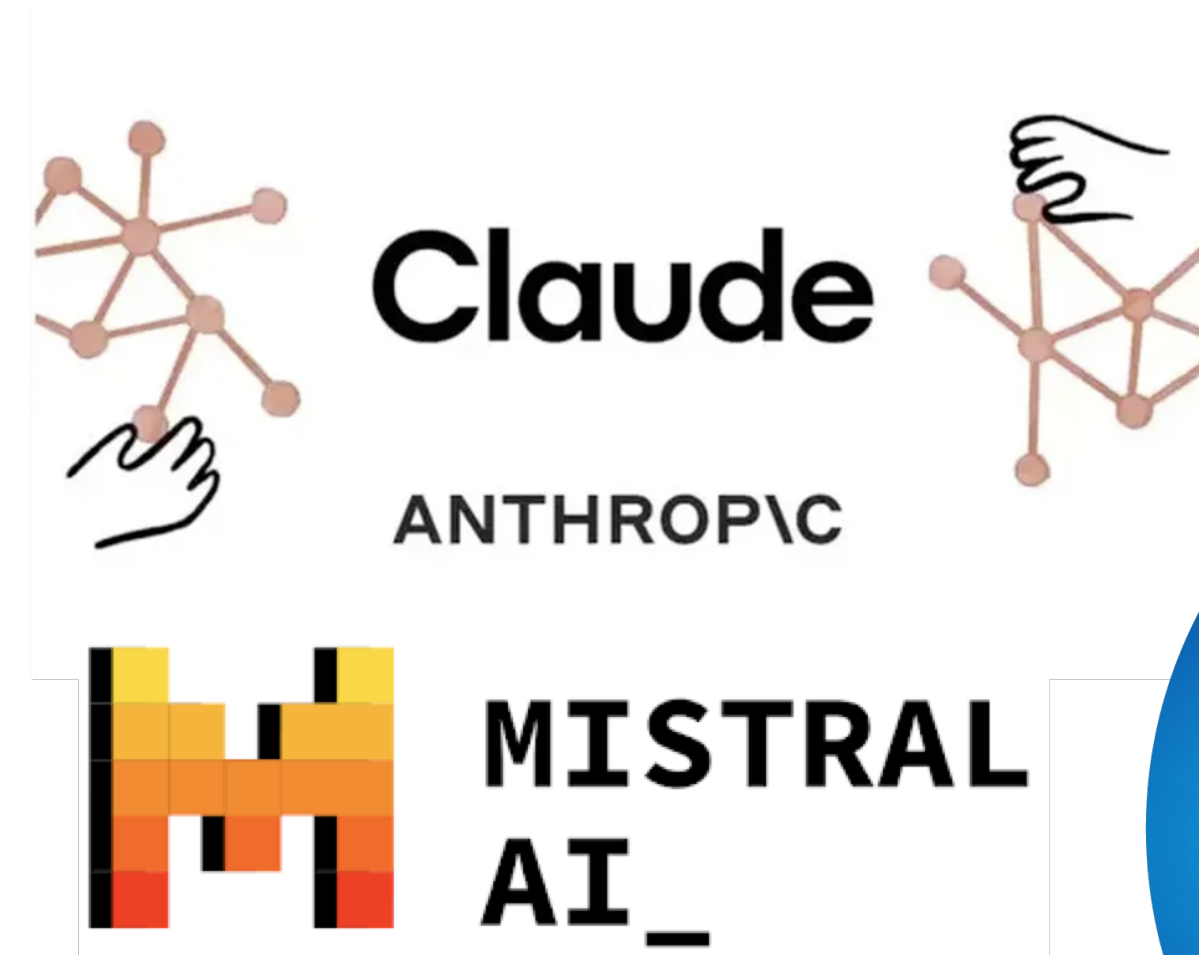
Linear Fit

Green Available Samples

Blue Training Steps

Red Tunable Parameters

$y=x$



Example application: Health data is sensitive and legally protected

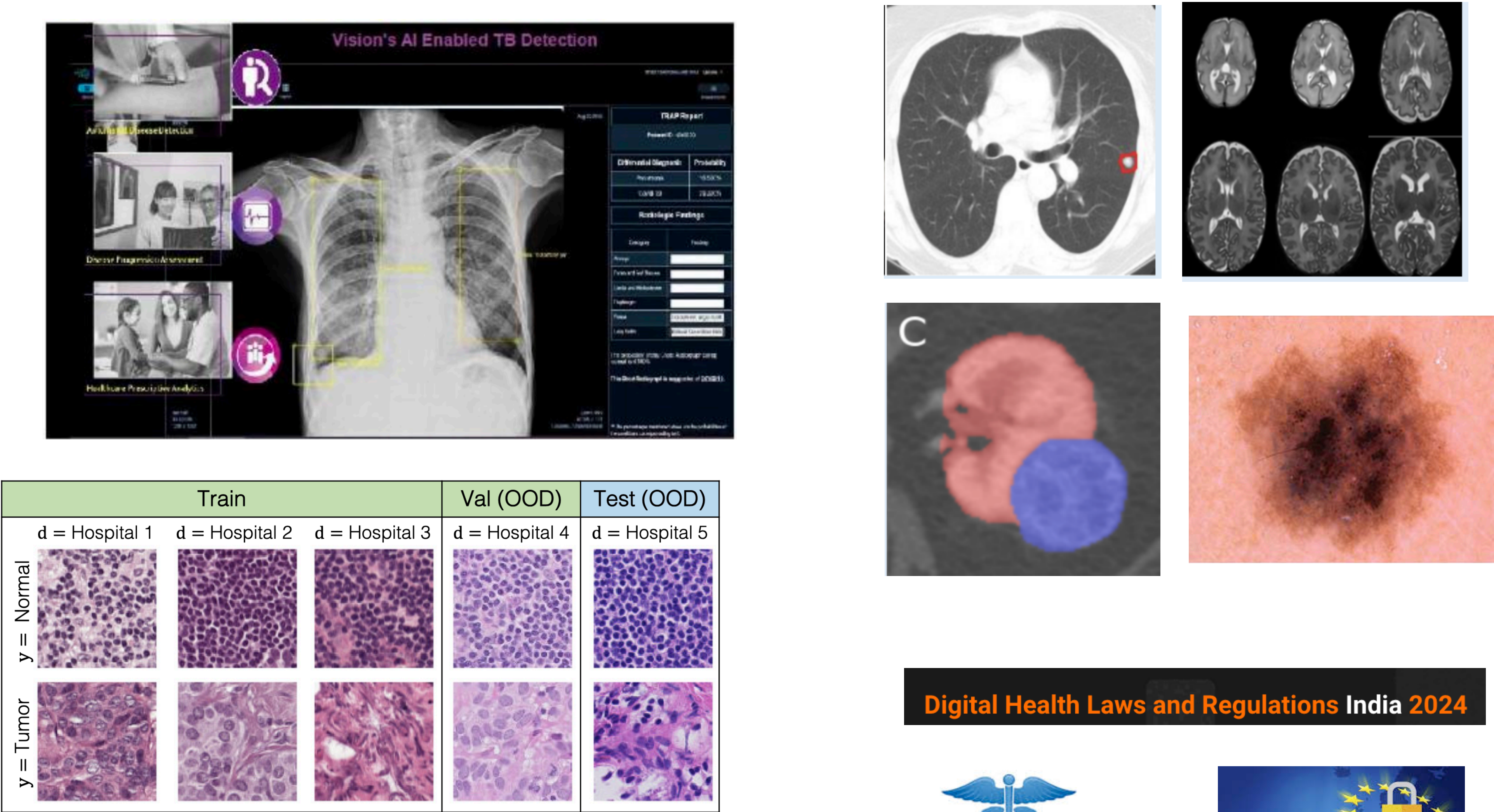
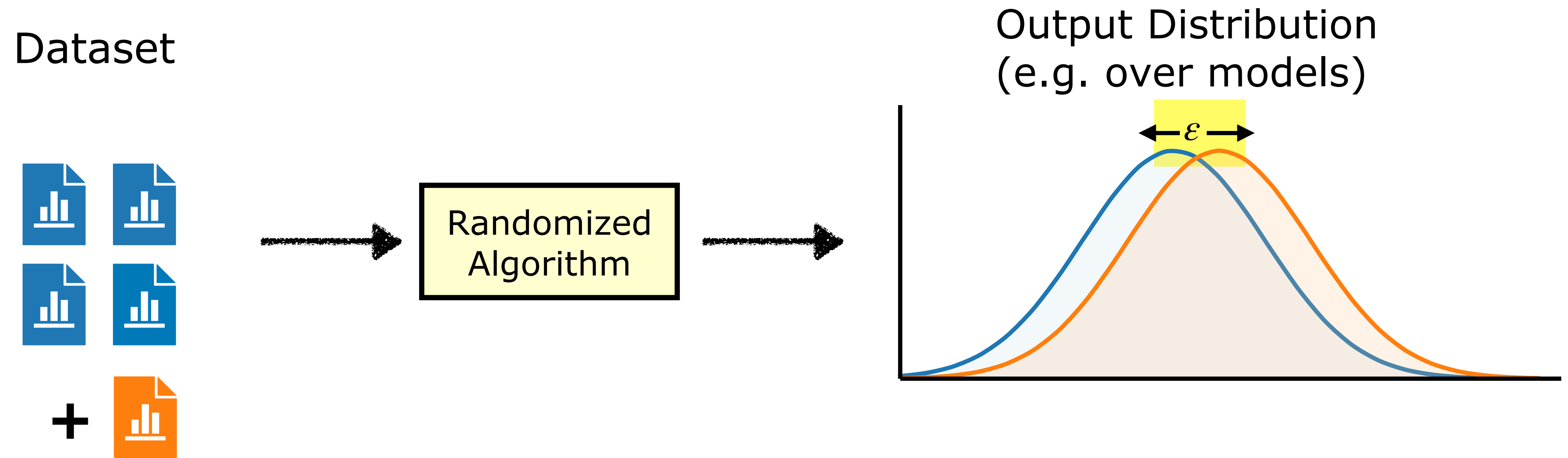


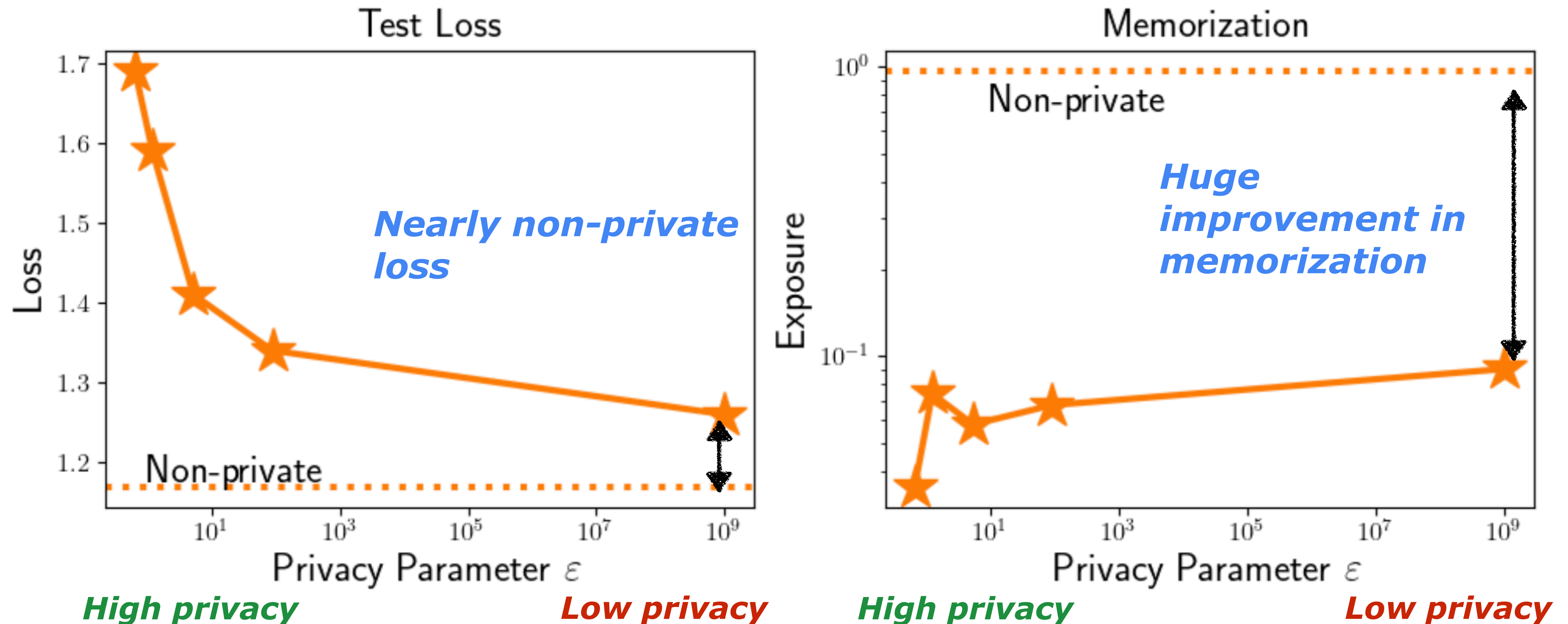
Figure 4: The CAMELYON17-WILDS dataset comprises tissue patches from different hospitals. The goal is to accurately predict the presence of tumor tissue in patches taken from hospitals that are not in the training set. In this figure, each column contains two patches, one of normal tissue and the other of tumor tissue, from the same slide.

Differential privacy nearly eliminates memorization



A randomized algorithm is ϵ -**differentially private** if the addition of **one user's data** does not alter its output distribution by more than ϵ .

Differential privacy nearly eliminates memorization



Carlini, Liu, Erlingsson, Kos, Song. **The Secret Sharer: Evaluating and Testing Unintended Memorization in Neural Networks.** USENIX Security 2019.

Today's Outline

- Logistics
- Course Outline

Logistics

Classes

- **Monday:** 5 to 5:50 pm
 - Mini lab/tutorial session. Will be announced in advance
 - Please **bring your laptops**. Contact us privately if you have strong impediment (e.g. cannot afford a laptop, etc.)
- **Wednesday:** 2 to 3:15 pm
 - Lecture, usually on the board
- **Thursday:** 3:30 to 4:45 pm
 - Lecture, usually on the board

Communication

- **Course Webpage:** https://krishnap25.github.io/privAI_course_2026e
- **Piazza:** link to be announced
- **Assignments:** submit via Moodle/Gradescope: TBD
- Do ***not*** contact us via email. Use piazza only
 - Public post: use for logistics, assignments, etc.
 - Private (instructor + TAs): For special requests

Grading

- ***Homework + Mini-quiz***: 30% (best 3 out of 4, 10% each)
- ***Course Project***: 40%
- ***Midterm exam***: 30%

Homework + Mini-Quiz: 30% of the grade

- **HW0**: pre-requisites
 - Not graded but submission is necessary to continue in the course
 - Out today, due Jan. 30th
- **HW1-4**: approx. 2 weeks days per assignment (**total score: 10% each**)
 - HW1: 1st half of Feb.
 - HW2: 2nd half of Feb.
 - HW3: 1st half of March
 - HW4: 2nd half of March

Homework + Mini-Quiz: 30% of the grade

- Grading of HW:
 - Theory part: ungraded, but we will have a **mini-quiz** in class after deadline
 - Programming part: graded as per usual
- HW+Mini-quiz = 10% (split varies depending on theory-programming split)
- Grade = best 3 out of 4

Course Project: 40% of the grade

- Most of your learning will be through the course project
- Groups of 2-3 (exact details TBD, depending on the final class size :)
- ***Options:***
 - Research project
 - Implementation: benchmarking and open-sourcing
 - In-depth paper analysis

Course Project: Research

- **Original research:** can be theory, applied, or mix or both
- Commensurate to a workshop paper at NeurIPS/ICML/ICLR conferences

Course Project: Research

- You can propose your own course project related to your research
 - Must be related to the course contents
 - E.g. You work in computer vision for healthcare:
Implement private training or privacy attacks etc. on your model/dataset
- We will also provide some project suggestions

Course Project: Implementation

- Implement existing algorithms with a goal of:
 - Benchmarking methods (e.g. compare to various baselines)
 - Creating or contributing to open-source packages



Opacus

Train PyTorch models with Differential Privacy

JAX-Privacy: Algorithms for Privacy-Preserving Machine Learning in JAX

[Installation](#) | [Reproducing Results](#) | [Citing](#)

Course Projects: In-depth paper analysis

- Read and analyse the results of a theoretical research paper
- Reproduce the proofs in your own words (possibly using alternate methods)

Course Project (40%): Logistics

- **Proposal (5%, 1-2 pages):** Due around mid-March (TBD, based on exam)
 - What do you want to do? Why? Identify important references.
- **Midpoint report (10%, 4 pages):** Due around early April
 - Literature review (explain ideas mathematically, as in lectures)
- **Presentation (10%):** End of April/early May (last week of classes)
- **Final report (15%, 6-8 pages):** Due around end sem week

Course Goals

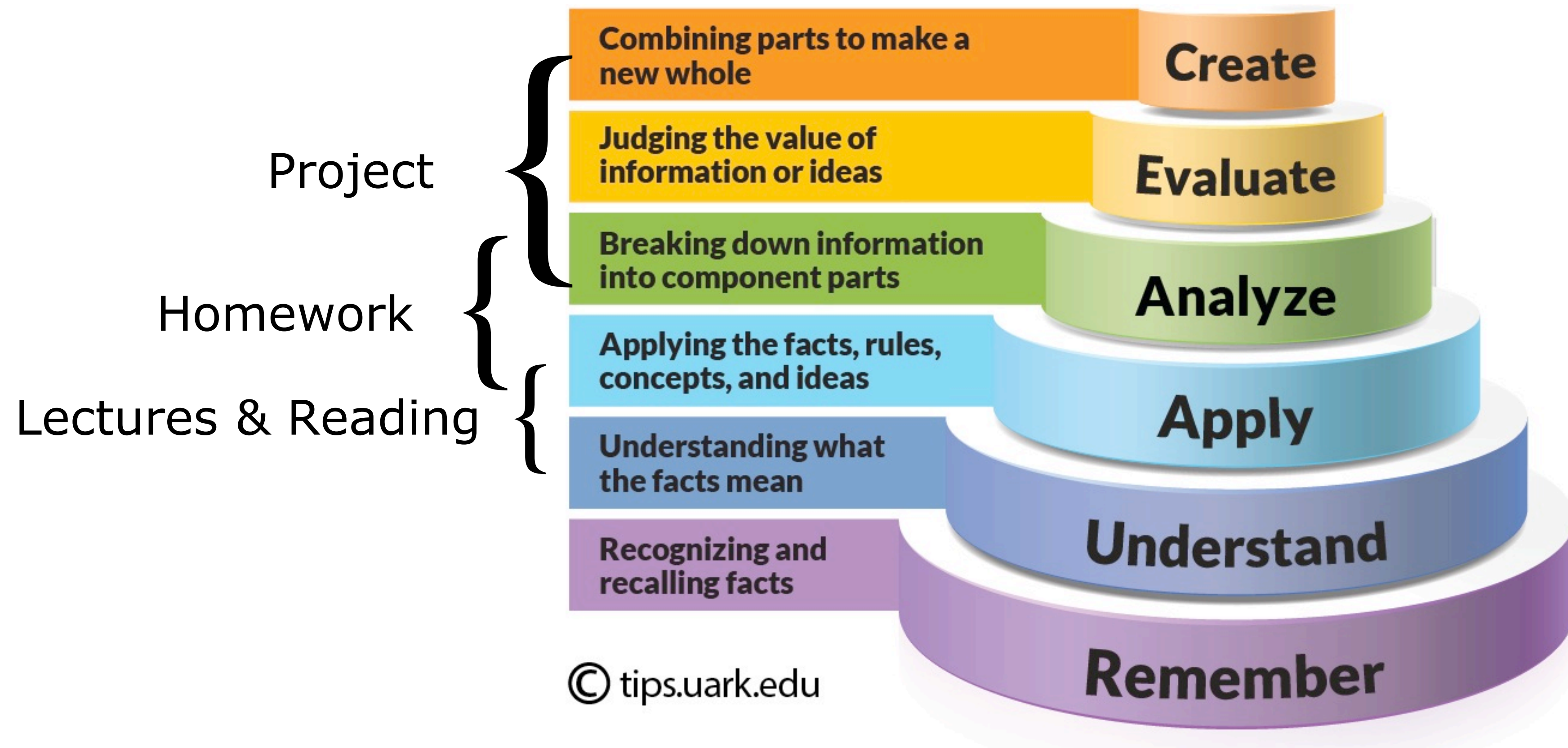


Figure: <https://tips.uark.edu/using-blooms-taxonomy/>

Bloom's Taxonomy

Midterm (30%)

- Duration: 3 hours
 - similar to an end-sem but we'll finish it early so you can focus on projects
- We'll decide the date/time next week
 - Approx. 2nd half of March (around quiz 2 time)

Late Days

- **No late days for project-related deadlines**
- HWs: You have a budget of 3 late days for programming assignments
 - No questions asked, you do not need to contact us for it
 - Even two minutes after the deadline is considered as a full late day
- Any further late days after exhausting your late day budget results in a zero grade for that homework

Honour Code

- **Project:**

- You have to do the work yourself (cannot use somebody else's work as yours for a course project)
- The project cannot be used "as is" for other courses
- Ok to reuse course project for BTP/DDP/MTP/other research projects
- Academic violations will be handled by the IITM Senate Discipline and Welfare (DISCO) Committee.

Honour Code (for HW submissions, e.g. programming assignments)

- **Homeworks:** Can collaborate with others to discuss homework, provided:
 - You acknowledge everybody you worked with in your submission
 - You write your own code independently without referring to written notes and other material from joint discussions
 - You must internalize the solution/code well enough to reconstruct it fully by yourself
 - ***Copy-pasting from each other is strictly not allowed***

Honour Code: LLM policy

Ok to use LLMs for HW/project but each time you do, you must provide:

- name of the LLM/co-pilot and the exact prompt (verbatim)
- A summary of how it helped

You are ultimately responsible for the answer you submit:

- You need to **understand what you submit**. We might test it and grade you on it
- It should be correct and sensible

- **Not allowed:** sharing your prompt verbatim is not allowed

Honour Code

- We expect and believe that you will conduct yourself with integrity
 - We will follow the institute policies but it is ultimately up to you to conduct yourself with academic and personal integrity for several compelling reasons (that go beyond your studies)
- **Respect diversity:** There is a place for everyone who is curious and passionate about exploring knowledge
 - Let us all be mindful of creating welcoming and inclusive spaces
 - As the next generation, you have the power to shape the future: aim to make the world a better place!

Office Hours

- We will be available one hour per week to answer queries about the course material
- **Wednesday 3:30 to 4:30 PM** at my office (after class)

Auditing the course

- **Very strongly discouraged**
 - Active learning works best: working on homework/project
- Students registered for credit will be given priority
 - Others may be requested to leave in case we run out of space

Attendance & Misc policies

- We will not take attendance
- This will be a challenging course. Take it only if you ***really*** want to learn the material
- Hard to follow without attending the lectures
- Please be on time. It is very disrespectful to be late
- No phones in the class please. Laptops for lab sessions only (usually Mondays)

Some course feedback from last offering

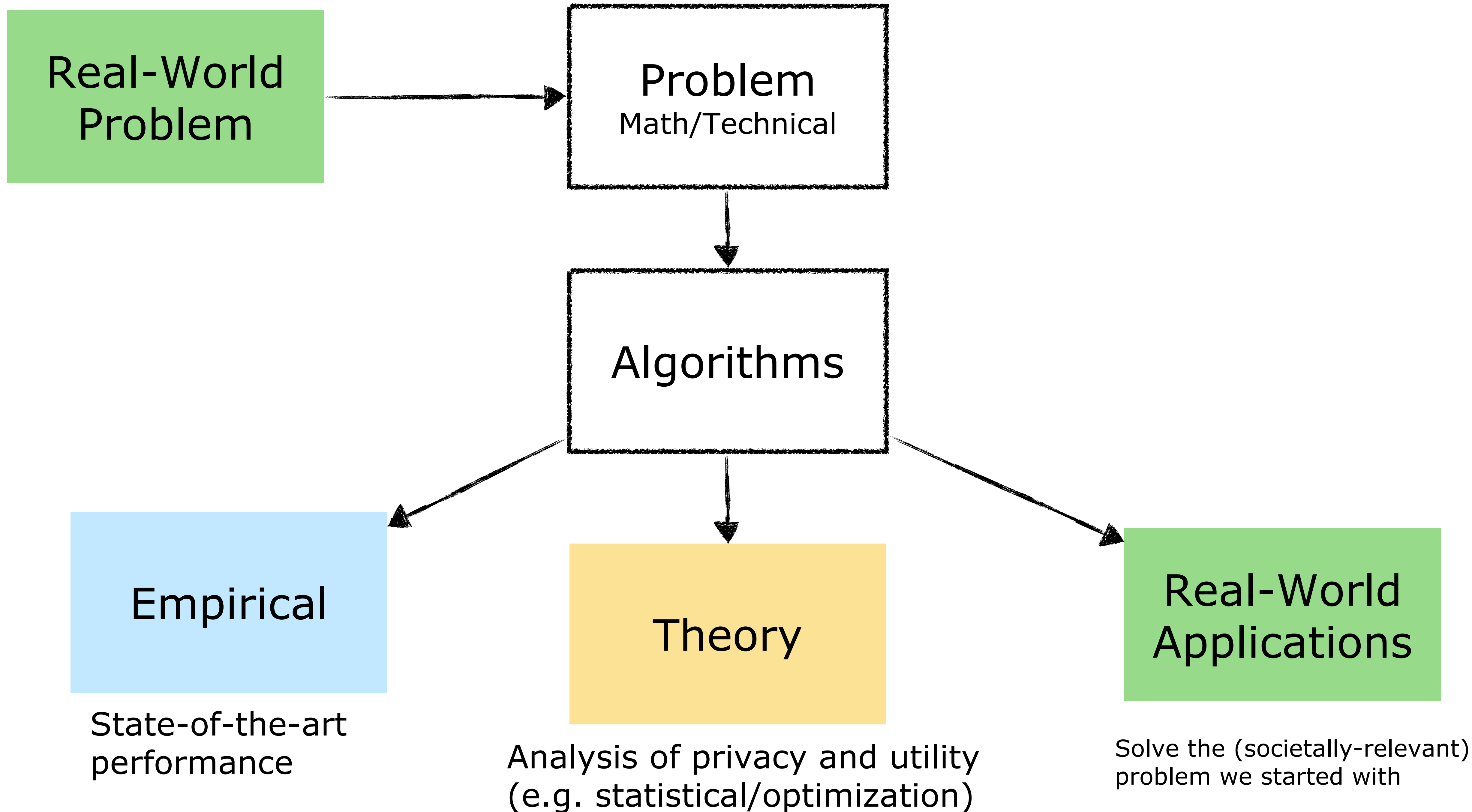
I would recommend them to take this course if they are interested in learning something really new and different in the field of ML. I would also caution them to have the required prerequisites for the course or else it would most likely not allow them to appreciate the course in itself.

Worth it if you put in the effort. Not an easy course though.

take it if you have no other commitments, and if you are interested in the topic.

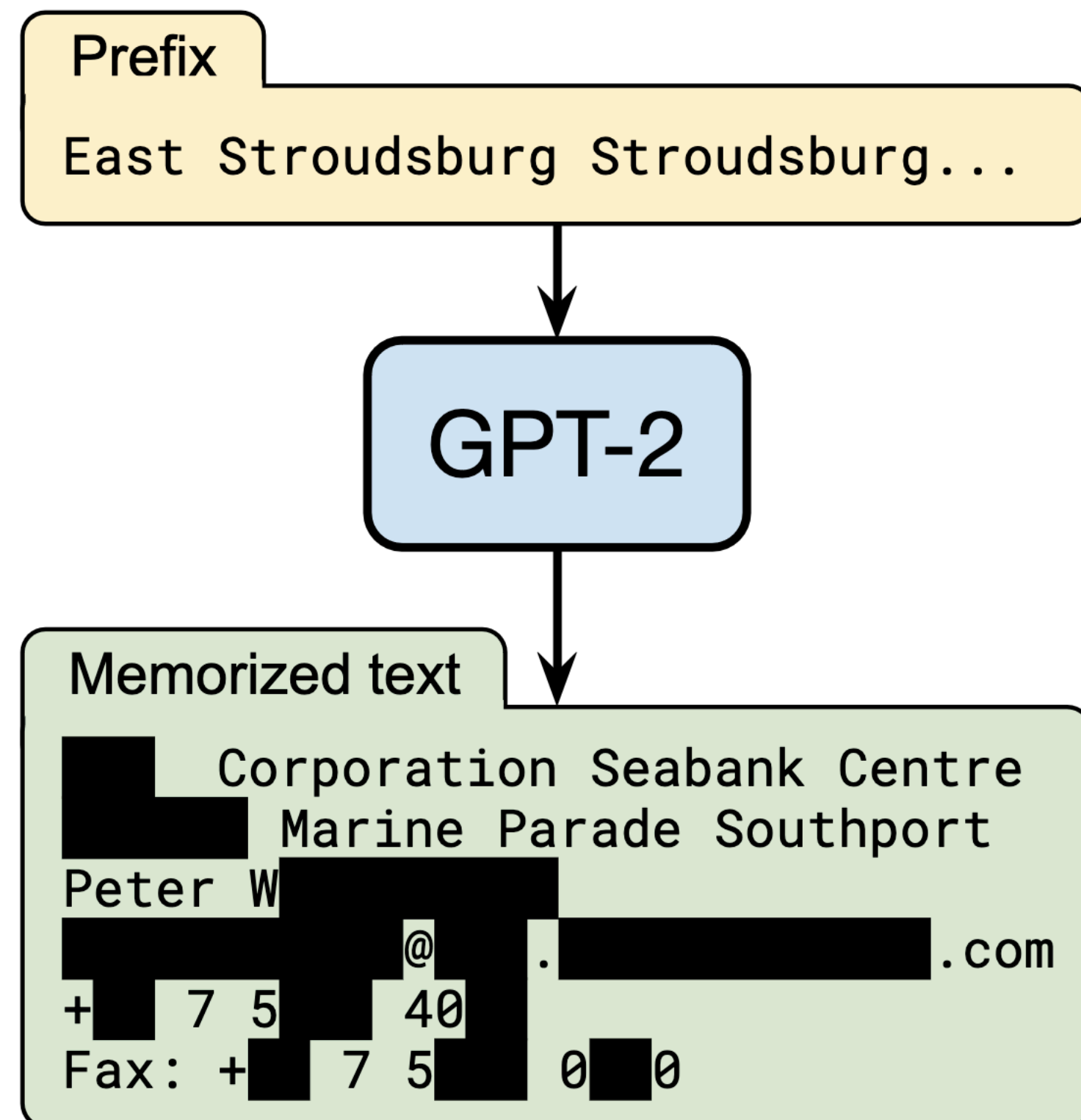
Highly recommended for students eager to delve deeply into mathematical concepts and cultivate a mindset for engaging with new research. However, I suggest that students carefully assess their workload before enrolling, especially if they are in a placement semester with additional courses and a thesis. Balancing these commitments can be extremely challenging.

Tentative Course Outline

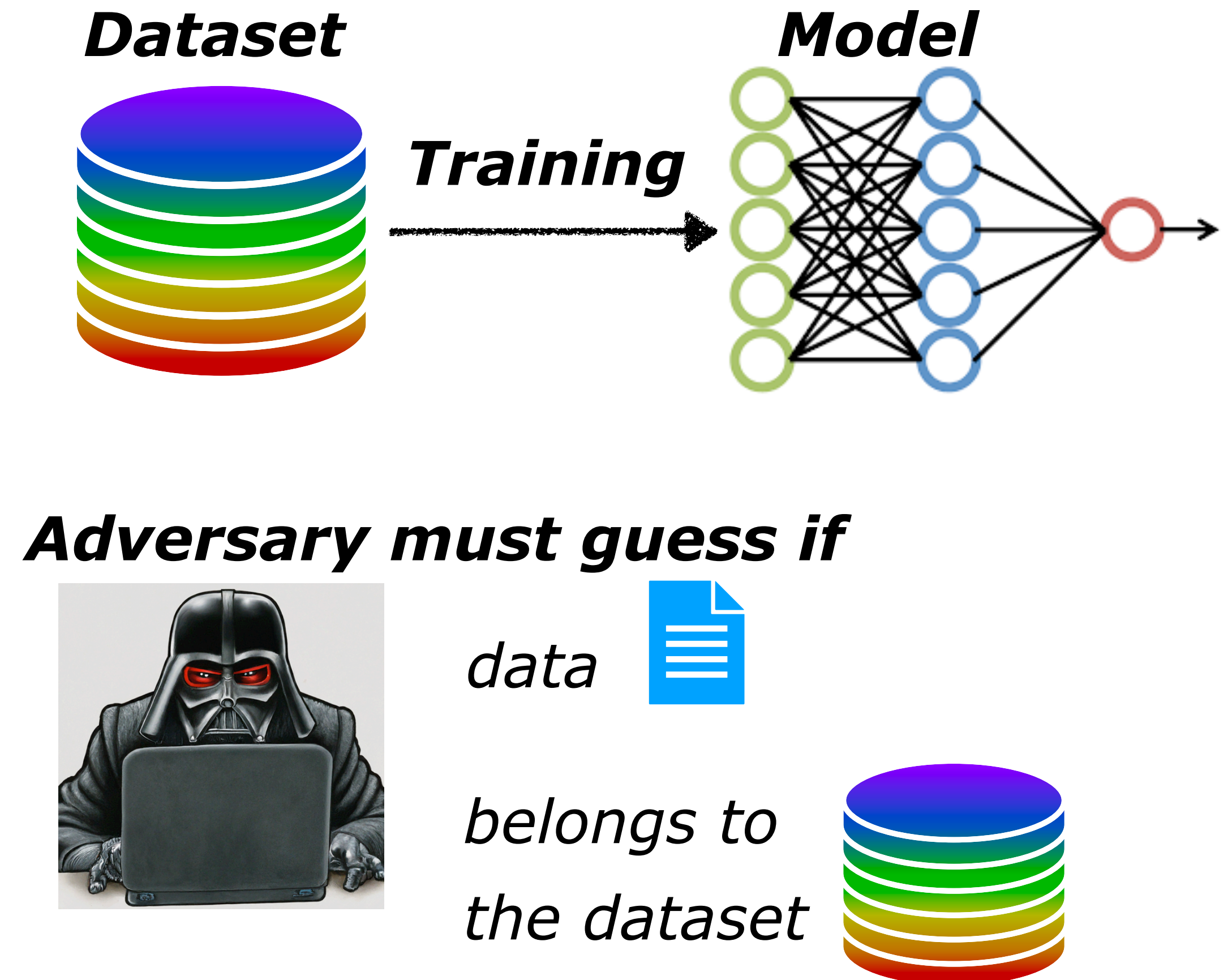


Week 1: Privacy Attacks

- Data Reconstruction/Extraction



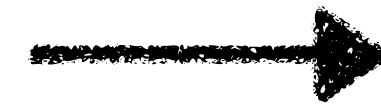
- Membership Inference



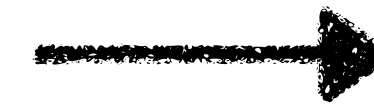
Weeks 2-4: Differential Privacy (DP)

A mathematically rigorous notion of “***privacy***”

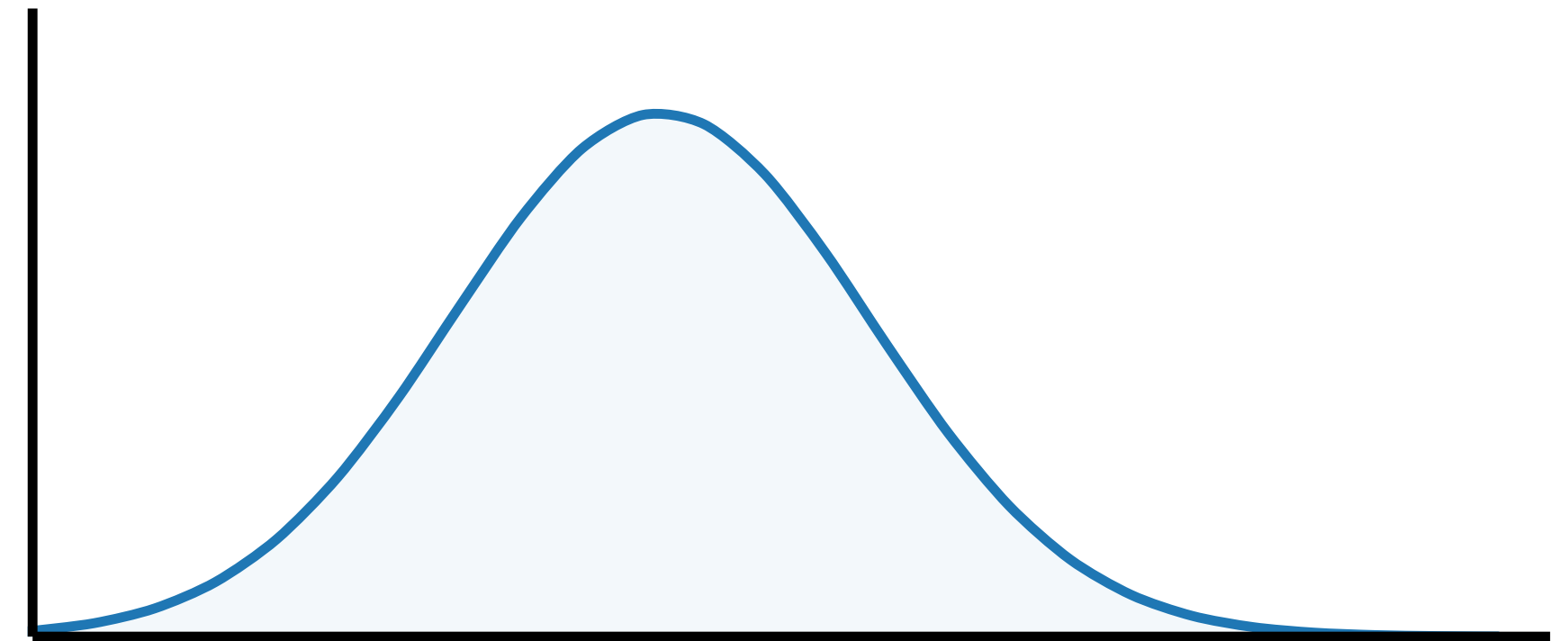
Dataset



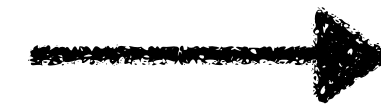
Randomized
Algorithm



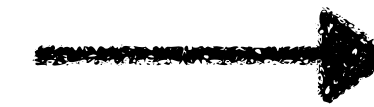
Output Distribution
(e.g. over models)



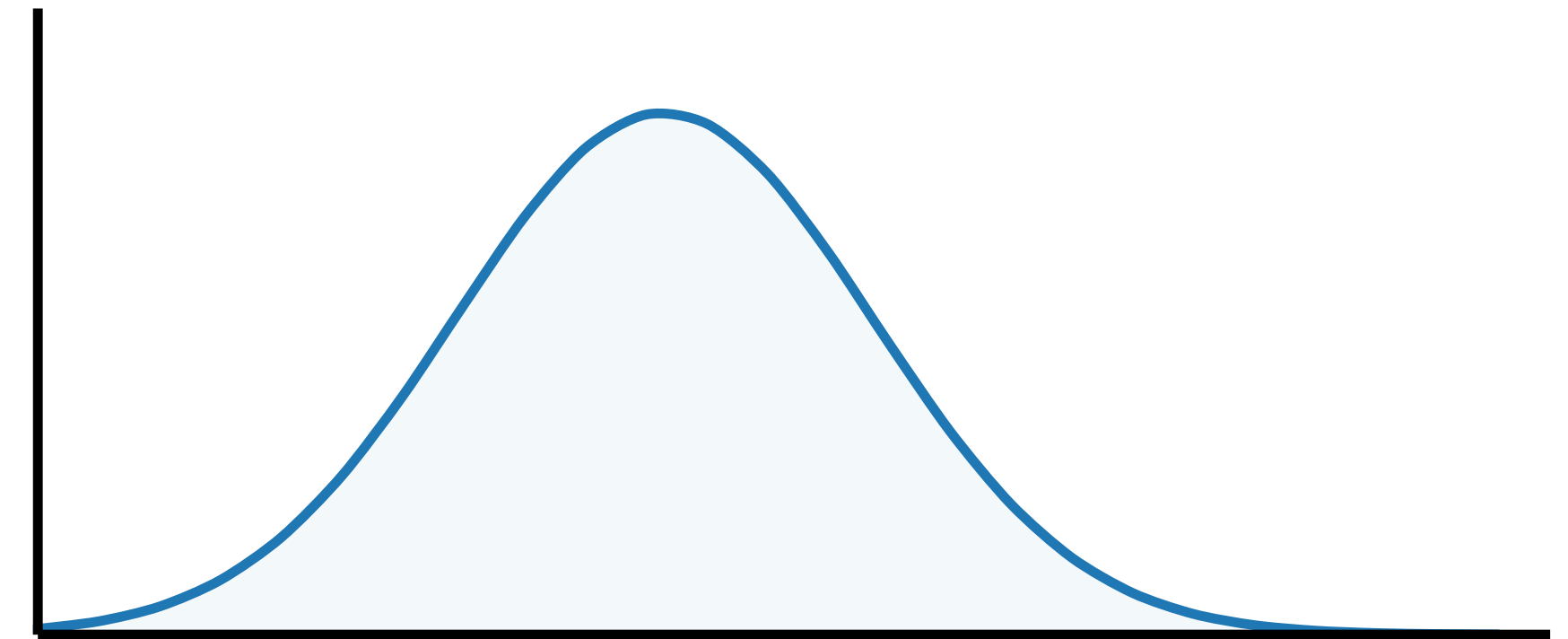
Dataset



Randomized
Algorithm



Output Distribution
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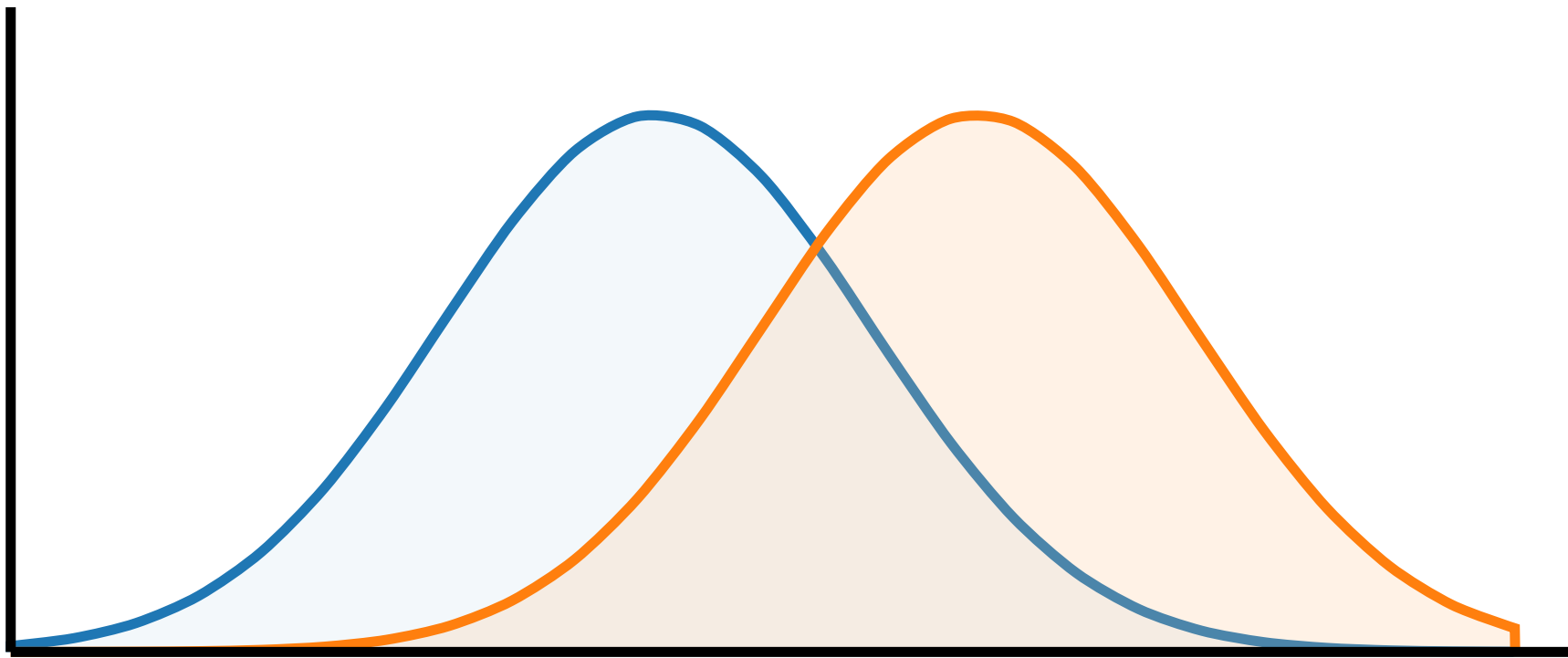
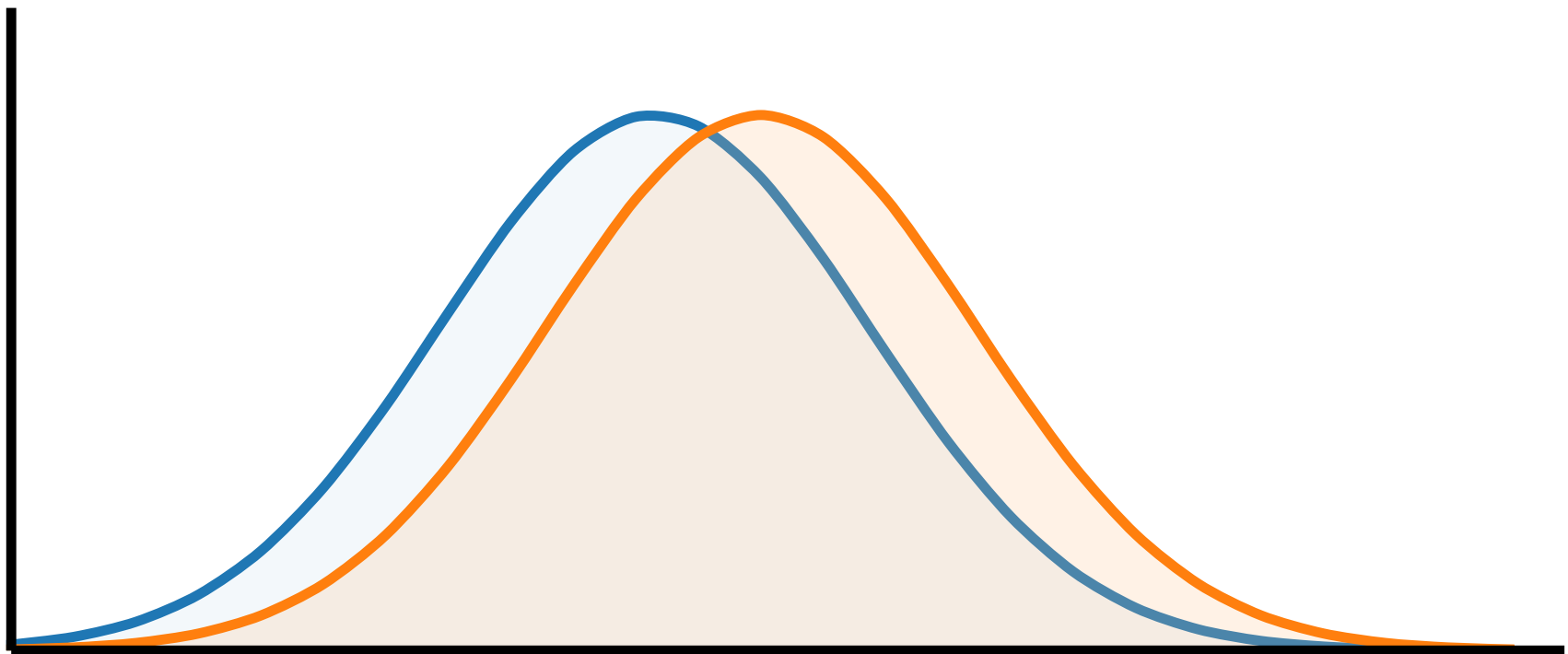
Dataset



Randomized
Algorithm



Output Distribution
(e.g. over models)

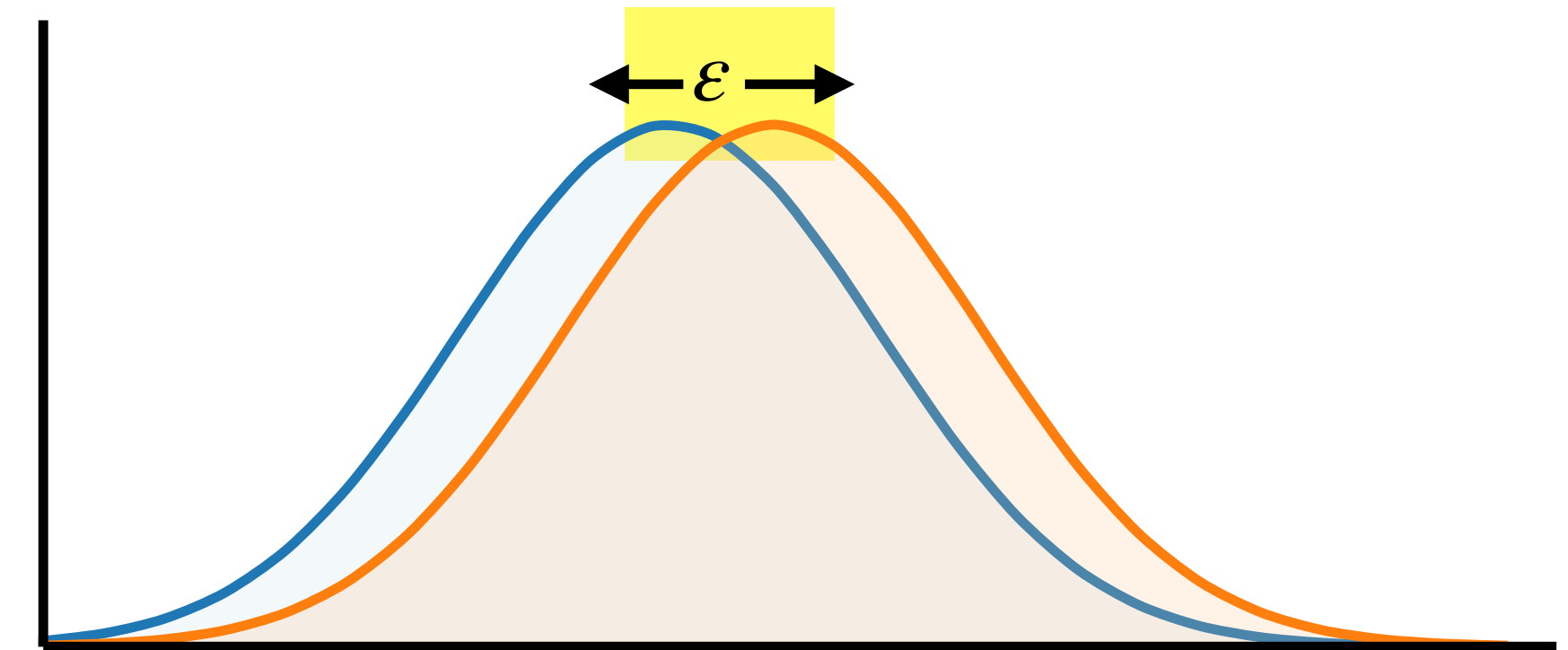


Dataset



Randomized
Algorithm

Output Distribution
(e.g. over models)

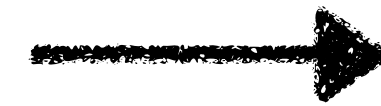


A randomized algorithm is ϵ -**differentially private** if the addition of **one user's data** does not alter its output distribution by more than ϵ .

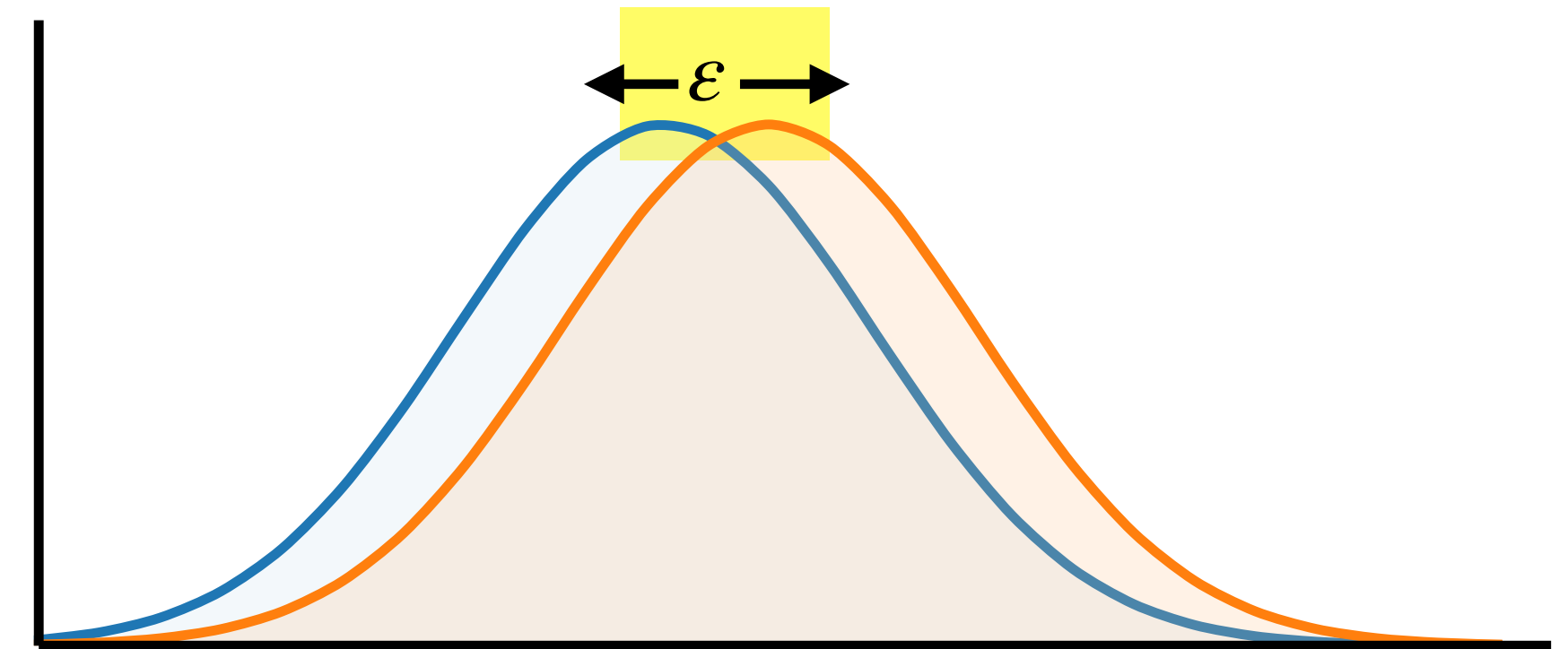
Dataset



Randomized
Algorithm

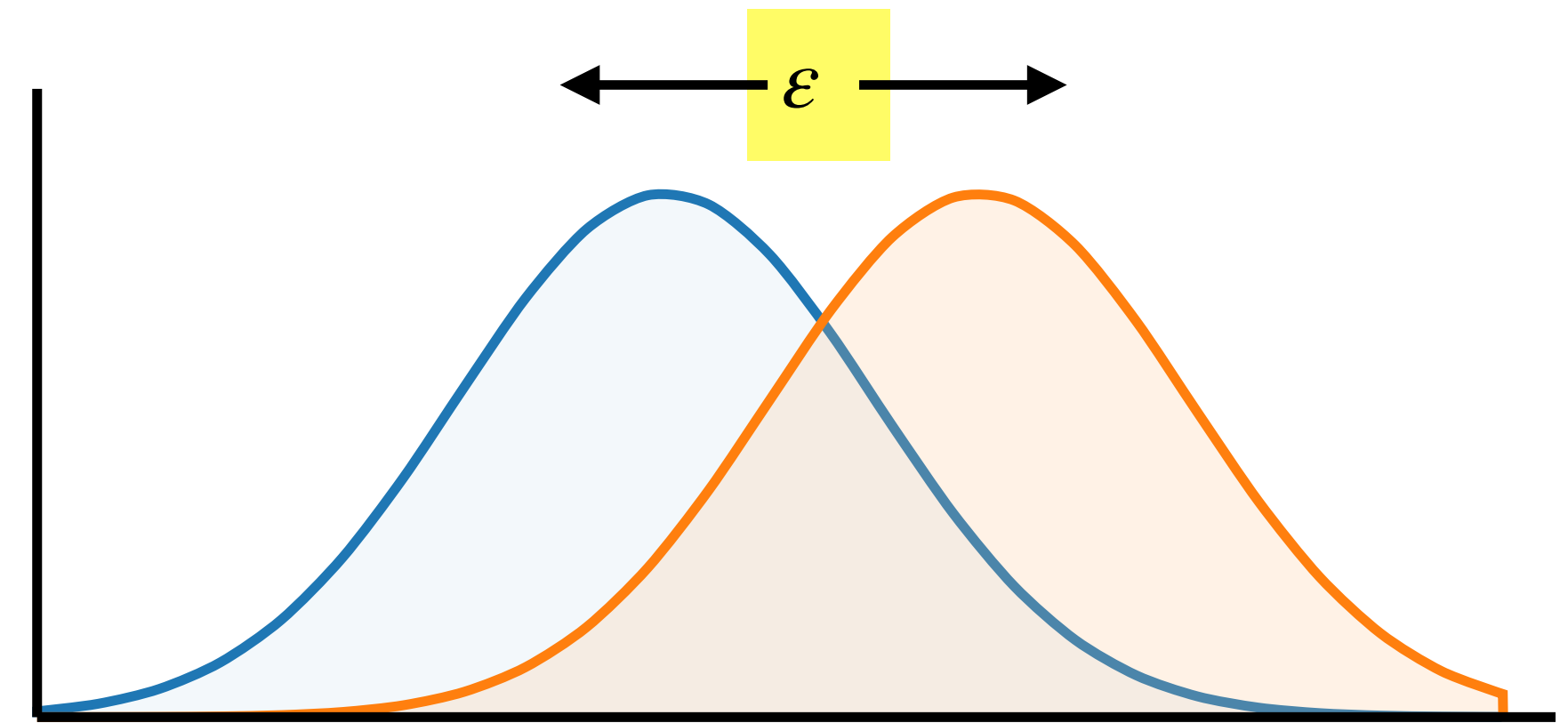


Output Distribution
(e.g. over models)



ϵ -differential privacy

Large $\epsilon \implies$ more privacy leakage



Caveat: Multiple facets of the word “***privacy***”

What does the word “***privacy***” mean to an end user of an AI product?



Transparency, Control,
Verifiability



Minimize data sharing



Data Anonymization

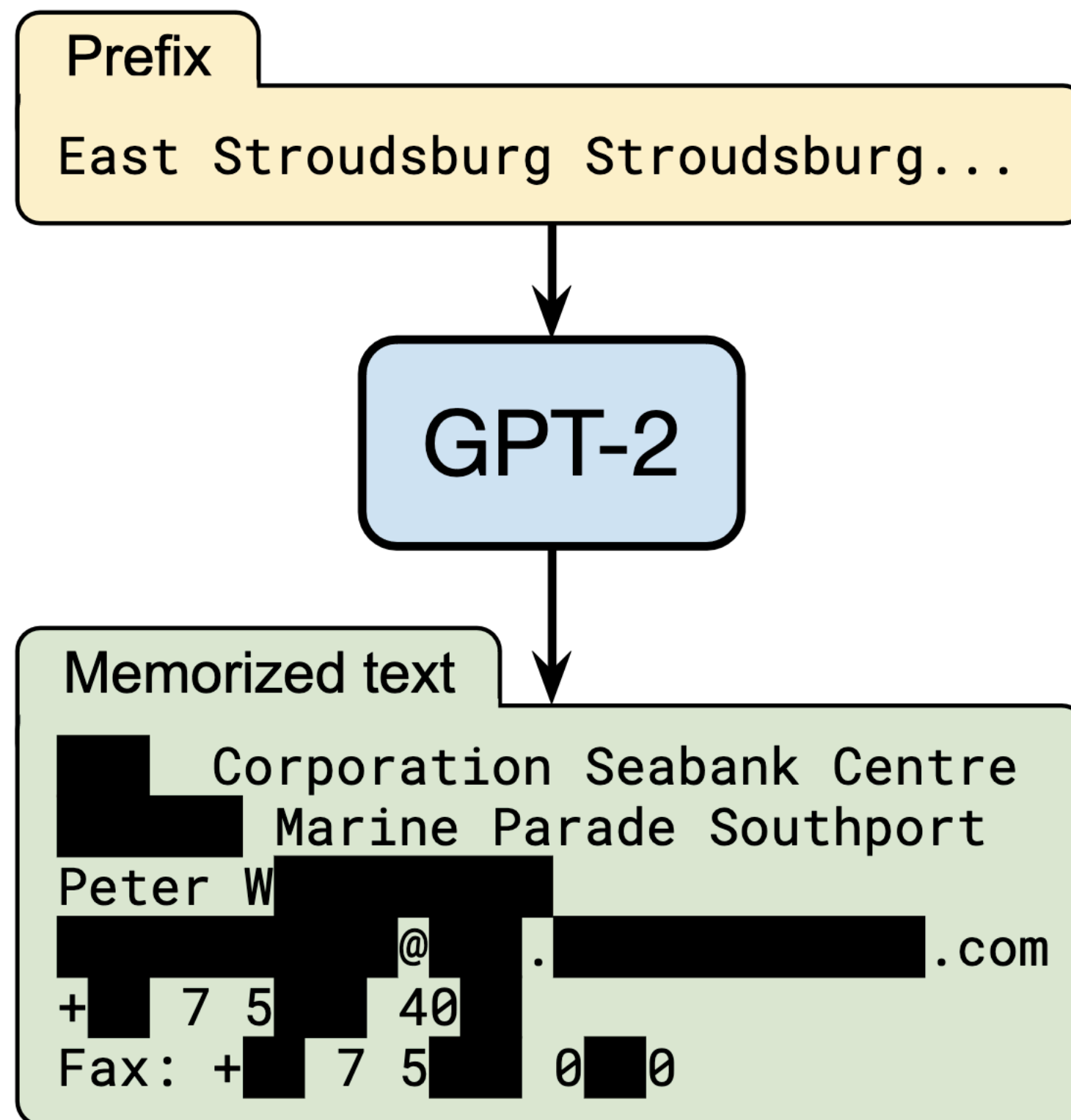
- *Differential Privacy*

Bonawitz, Kairouz, McMahan, Ramage (2022). **Federated Learning and Privacy.**
Communications of the ACM.

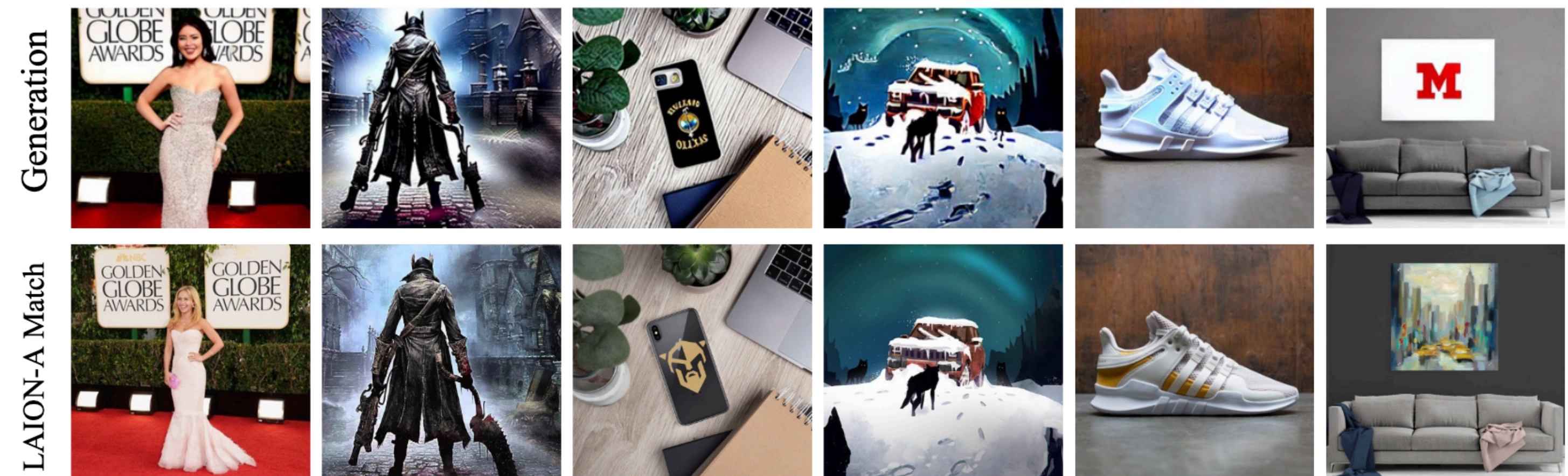
Weeks 5-7: ML with DP

Modifying stochastic gradient descent for DP

Weeks 8-9: Protecting Against Data Reconstruction Attacks



Generated Images



Real Images
(from training)

Weeks 10: Advanced Topics

- Privacy in distributed settings
- Generative AI:
 - Contextual privacy norms
 - Copyright
- Unlearning
- ***Your suggestions welcome***
- ***Guest speakers from industry and academia***

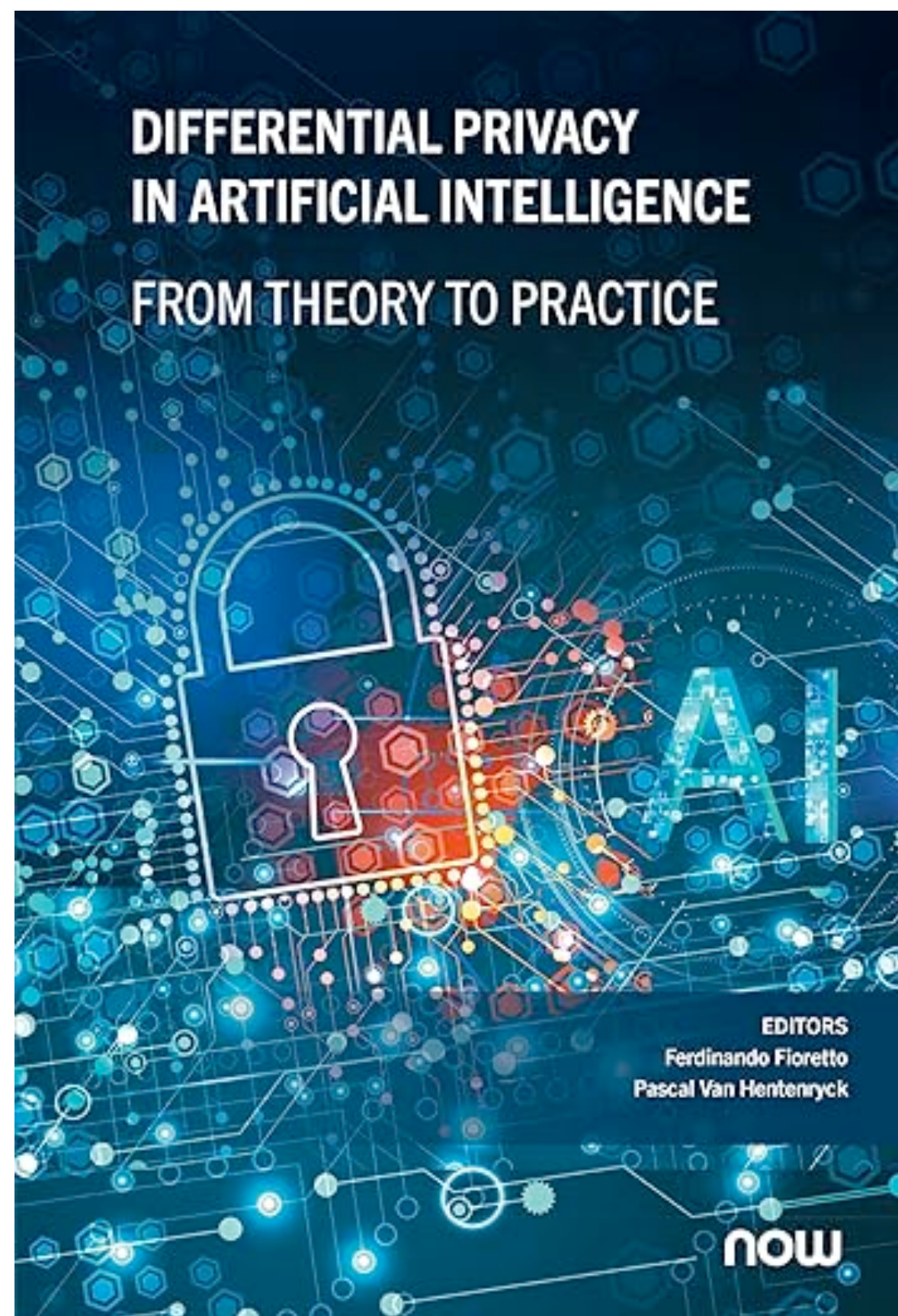
Last week of classes

- Project presentations

Reference Material

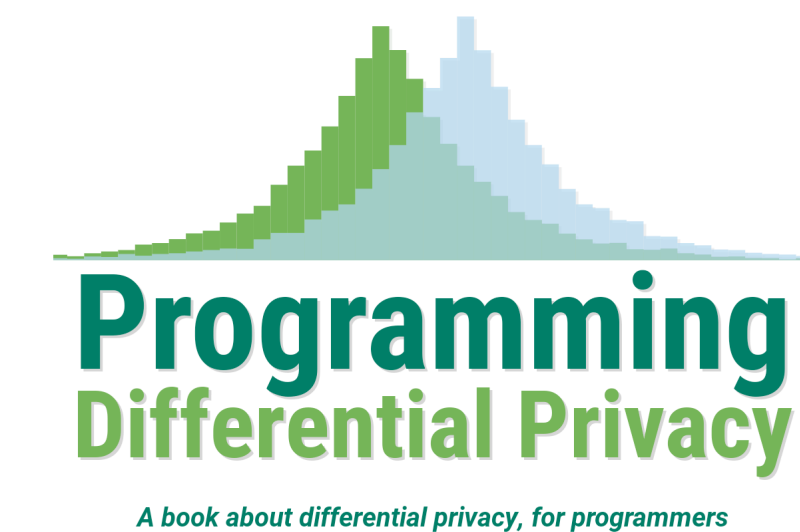
Main book

(free PDF available)



Supplementary reading:

- Papers: posted on course webpage
- Programming DP book: (available online for free)



Joseph P. Near
Chiké Abuah

Thank you! Questions?